



CS - Applications OF ICT

Lab Manual

Department of Software Engineering
Military College of Signals
National University of Science & Technology

PREFACE

This Lab Manual is designed for 14 lab sessions, comprising 12 structured experiments, one open-ended lab session, and a final lab examination. The experiments are tailored to **"Applications of ICT"** and aligned with the curriculum guidelines recommended by **PEC** and **HEC**. The labs focus on practical ICT applications, including basic programming, data handling, problem-solving, and the use of common ICT tools. **Python** is used throughout the course, with **Spyder, PyCharm, Jupyter Notebook, or Google Colab** providing practical environments for writing, executing, and testing programs.

The manual aims to reinforce theoretical concepts through hands-on implementation and develop students' essential programming and ICT skills.

Course Instructor: _____

Lab Instructor: _____

Version History

Date	Updated By	Details
September 2026	Senior Lecturer Fawad Khan/ LE Tasbiha Sohail	First Version Created

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Lab 01 Introduction to Computer Hardware & MS Office

Objectives:

The main objective of this lab is to provide students with fundamental knowledge of computer hardware components and their functions, enabling them to understand how different parts of a computer system work together. Students will also gain practical skills in using MS Word, focusing on creating, editing, formatting, and managing professional documents. Through hands-on practice, they will learn to apply various features of MS Word such as inserting tables, images, headers, footers, and page layouts to enhance the quality of documents. By the end of the lab, students will be able to identify and explain essential hardware components and demonstrate proficiency in preparing well-structured and formatted documents in MS Word for academic and professional purposes

Hardware/Software Tools:

Hardware: Desktop/Computer
Software Tool: MS Word

Lab Description:

Computer hardware and MS Word is fundamental for developing essential computing skills. Computer hardware provides the physical components of a computer system, such as the CPU, memory, storage devices, and input/output peripherals, which together enable efficient processing and management of data. Learning about these components helps students recognize how computers operate and how each part contributes to overall system performance. MS Word, on the other hand, is a powerful tool for creating and formatting documents, where students can practice typing, editing, and organizing text. Features such as tables, images, page formatting, and styles allow users to prepare professional and well-structured documents. This lab introduces students to both the basics of computer hardware and the practical use of MS Word, enabling them to build foundational skills in computer operation and document preparation for academic and professional use.

Computer System:

Computer systems are broadly divided into two main categories:

- Hardware
- Software

Hardware: Hardware refers to the physical components of a computer system that the user interacts directly or indirectly. These include input devices such as the mouse, keyboard, and touchpad, which serve as user interface devices. Over time, these devices have evolved significantly. For example, modern keyboards now include ergonomic layouts, multimedia controls, and programmable macro keys, while advanced mice include high-resolution optical sensors and customizable buttons for gaming and design applications.

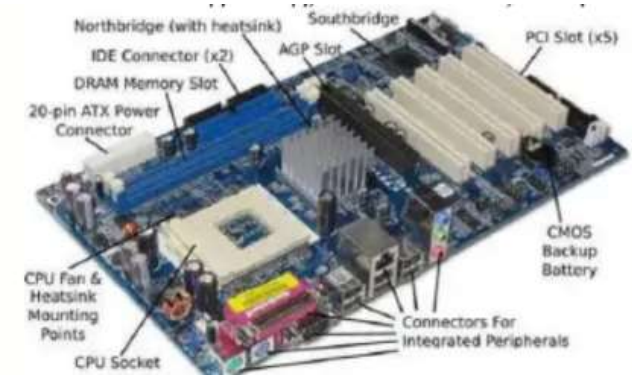
Software: Software provides the interface through which the user communicates with hardware. This can be through a Command Line Interface (CLI) or a Graphical User Interface (GUI). Historically, interaction was limited to punch cards and command lines, but today, modern operating systems and applications offer intuitive GUIs for user convenience and productivity.

Major Hardware Components of a Computer System:

Central Processing Unit (CPU): The CPU is the brain of the computer, responsible for executing instructions and performing arithmetic and logical operations. Modern CPUs, such as Intel Core i9, AMD Ryzen 9, and Apple M-series processors, contain multiple cores, hyper-threading, and integrated cache systems to handle complex and parallel workloads efficiently. CPUs are optimized for general-purpose computing, balancing performance and energy consumption.



Motherboard: The motherboard serves as the backbone of the computer system, connecting all components together. Modern motherboards include Pickle 5.0 slots, NVMe SSD interfaces, USB-C/Thunderbolt ports, and built-in Wi-Fi and Bluetooth modules. They also support overclocking, advanced cooling systems, and high-speed RAM modules, making them essential for gaming, AI workloads, and professional computing.



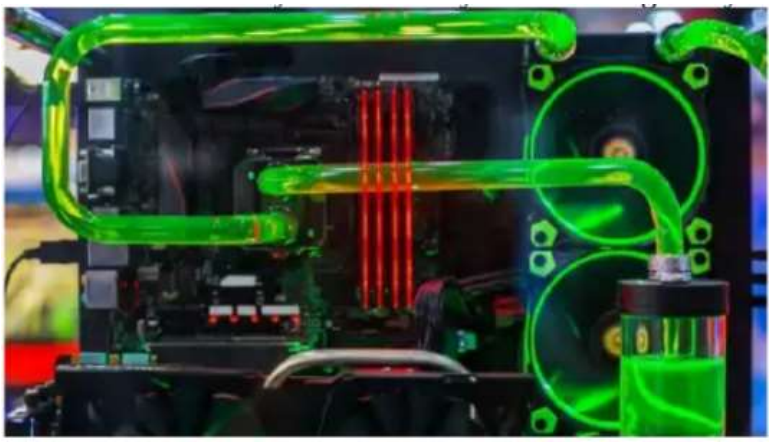
BIOS / UEFI Firmware: The BIOS (Basic Input/output System), now commonly replaced with UEFI (Unified Extensible Firmware Interface), initializes hardware during startup and provides a low-level interface for configuring system settings. UEFI allows faster boot times, secure boot features, and advanced hardware management compared to traditional BIOS.



Power Supply Unit (PSU): The power supply unit converts electrical power into a usable form for the computer's components. Modern PSUs are modular (detachable cables), energy-efficient (80+ ratings), and include power delivery systems for high-performance GPUs and multi-core processors.



Cooling Systems (Fans, Heat Sinks, and Liquid Cooling): Cooling is essential to prevent overheating. Alongside fans and heat sinks, liquid cooling systems and even immersion cooling technologies are used in modern high-performance PCs, servers, and AI data centers. This ensures stability and efficiency when running heavy workloads.



Ports and Connectivity: Modern computers feature a wide range of ports such as USB- C, Thunderbolt 4, HDMI 2.1, and DisplayPort 2.0, which support high-speed data transfer and high-resolution displays. Wireless connectivity has also advanced with Wi-Fi 6E and Bluetooth 5.3, enabling faster and more reliable connections



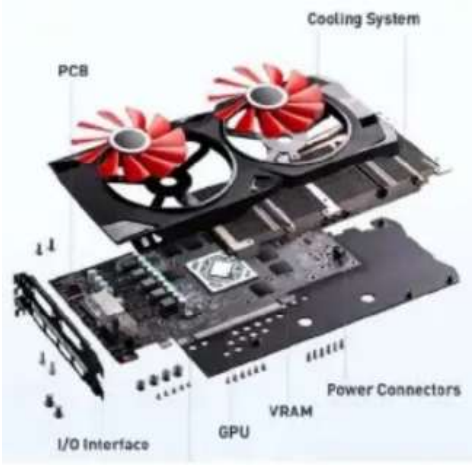
Internal Memory (RAM): RAM (Random Access Memory) is essential for temporary data storage during processing. Modern RAM technologies include DDR5 memory, offering higher speeds, better power efficiency, and improved multitasking capabilities compared to previous generations.



Storage Devices: Storage has evolved from traditional HDDs to Solid State Drives (SSDs), particularly NVMe SSDs, which provide lightning-fast read/write speeds. Cloud storage and hybrid systems are also widely used in modern computing.



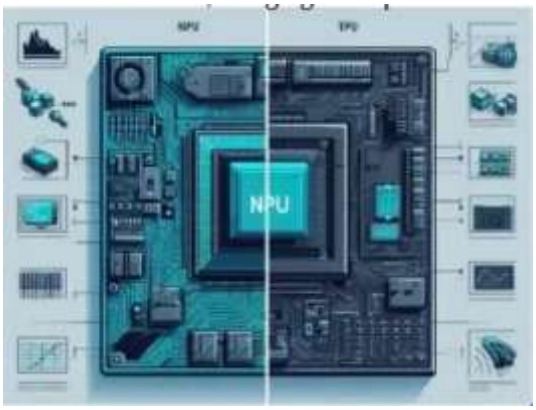
Graphics Processing Unit (GPU): A GPU is a specialized processor designed for parallel processing, originally for rendering images and videos. Modern GPUs, such as NVIDIA RTX and AMD Radeon, are crucial for AI, machine learning, deep learning, scientific simulations, and gaming. They provide massive parallel computation power and often work alongside CPUs.



Tensor Processing Unit (TPU): A TPU is a hardware accelerator developed by Google, specifically designed for machine learning and deep neural network processing. Unlike GPUs, TPUs are optimized for tensor operations, making them highly efficient for AI workloads, natural language processing, and computer vision applications.



Neural Processing Unit (NPU): NPUs are specialized chips designed to handle AI inference tasks efficiently. Found in modern smartphones (e.g., Apple Neural Engine, Qualcomm Hexagon), NPUs accelerate tasks like speech recognition, image enhancement, and real-time translations, bringing AI capabilities to everyday devices.



Application-Specific Integrated Circuits (ASICs): ASICs are custom-built chips designed for a specific application, such as cryptocurrency mining, AI model training, or networking. They provide unmatched speed and efficiency for their intended task but lack flexibility for general computing.



MS Word:

MS Word, a word processing tool, can be used effectively for data analytics by creating structured reports. With its features such as tables, formatting options, and the ability to insert charts and graphs from Excel, Word allows users to present data clearly. We can also perform basic data analysis, such as sorting and filtering, to gain insights from data. Key features of data analytics include:

1. **Tables:** We can insert tables to organize and structure data. For example, we can create a table that lists employee's names, their performance scores, and feedback ratings. Click on insert and select rows and columns according to the requirement.

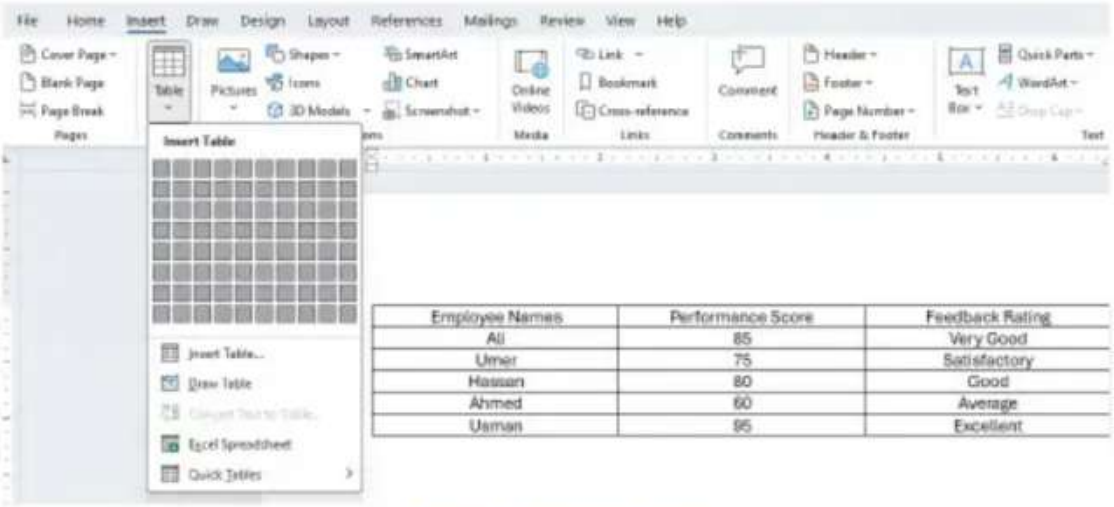


Figure 1: Inserting Tables

2. **Sorting/Filtering**: We can sort and filter data in tables to find specific patterns. For example, we can sort employee performance data by highest to lowest scores. In the Home tab and in paragraph section, select the sort option and select the column you want to sort.

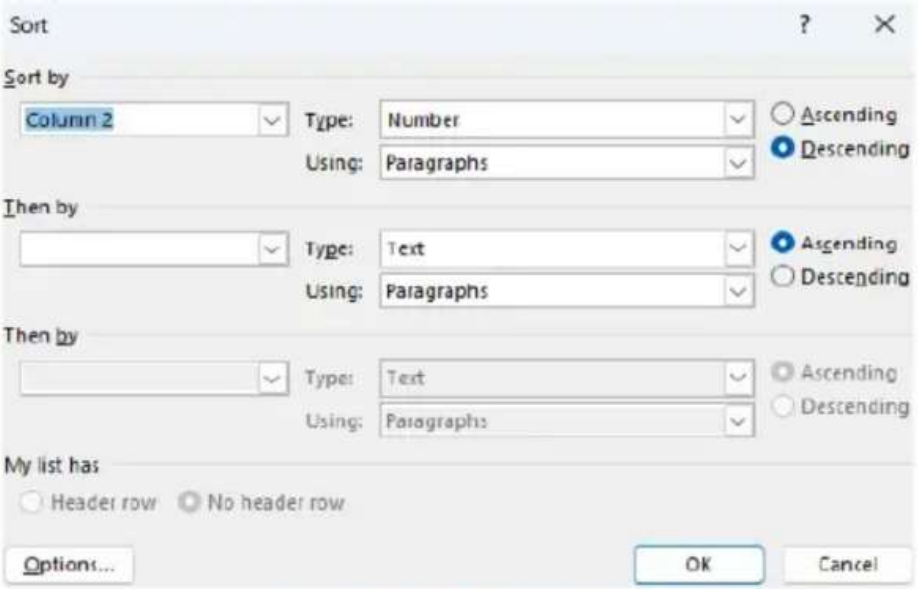


Figure 2: Sorting the data given in tables

Employee Names	Performance Score	Feedback Rating
Usman	95	Excellent
Ali	85	Very Good
Hassan	80	Good
Umer	75	Satisfactory
Ahmed	60	Average

Figure 3: Sorted Table

3. **Document Formatting**: Documents formatting includes features like headings, bullet points, numbered lists, borders, headers and footers, page numbers, page colors, water marks etc. In the Home tab, Reference tab and Design tab we have all these options.



Figure 4: Inserting different heading styles

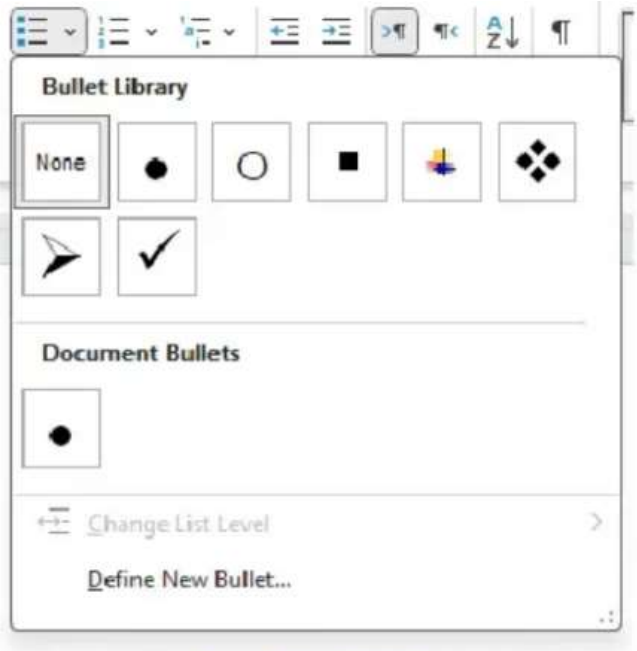


Figure 5: Inserting Bullets

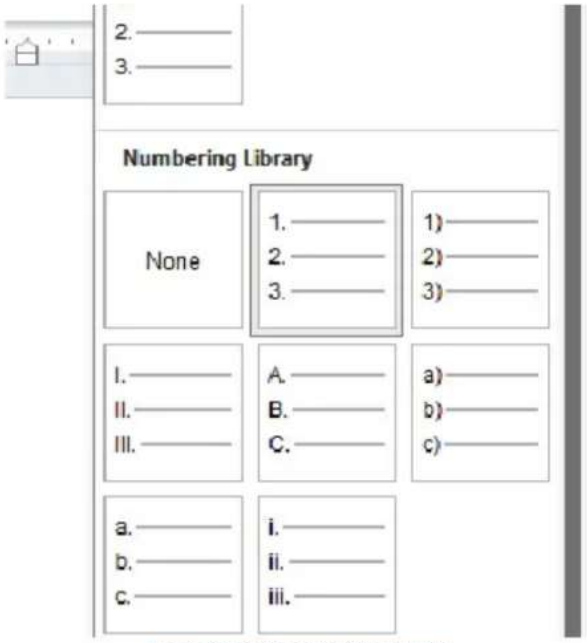


Figure 6: Inserting Numbered Lists

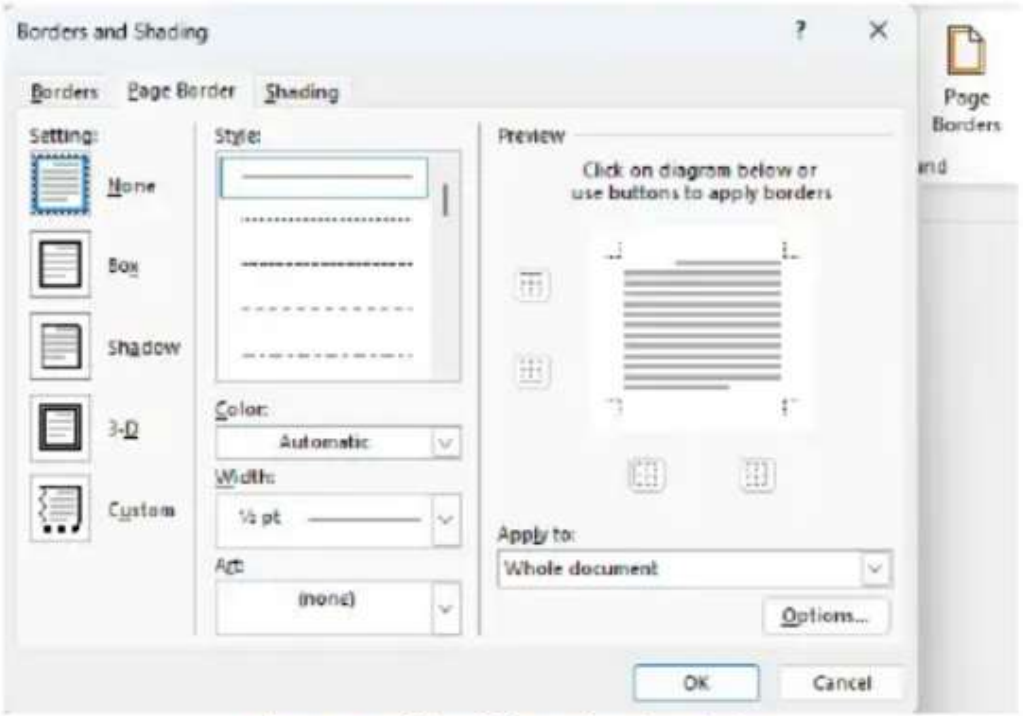


Figure 7: Adding different borders design

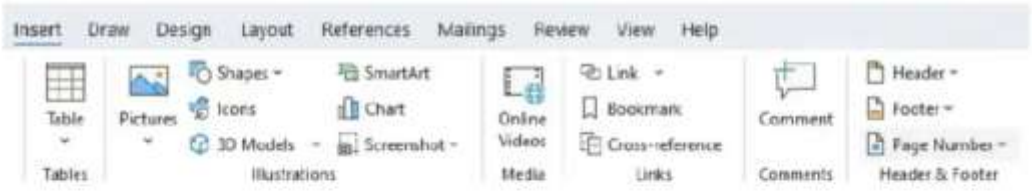


Figure 8:Inserting header, footer & page numbers



Figure 9: Adding colors to the page



Figure 10: Adding Watermarks

4. **Charts in Microsoft Word**: Charts are important for visualizing and presenting data in a clear, concise manner. They allow users to represent data in graphical form, making it easier to analyze trends, patterns, and comparisons by using bar charts, pie charts, line charts, and column charts, each serving different purposes for data representation. In the insert tab, select the option chart and insert the chart of your own choice. We can also update the data of chart just by clicking on the chart.

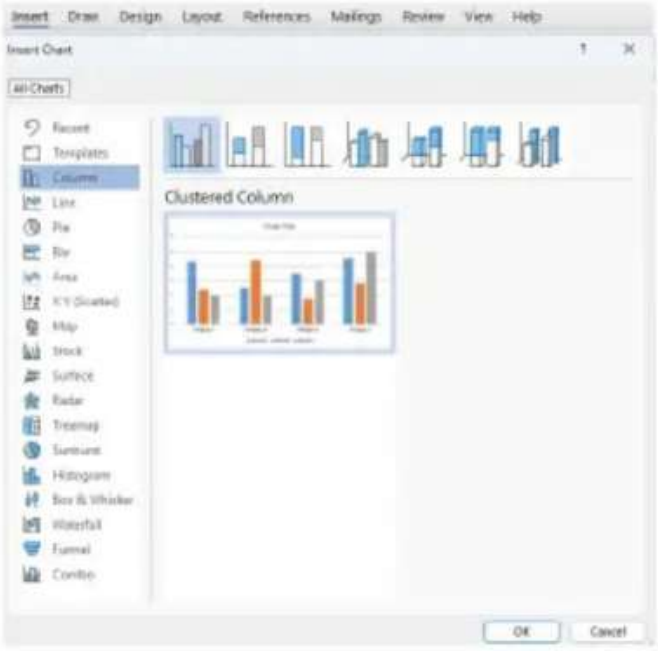


Figure 11: Inserting Chart

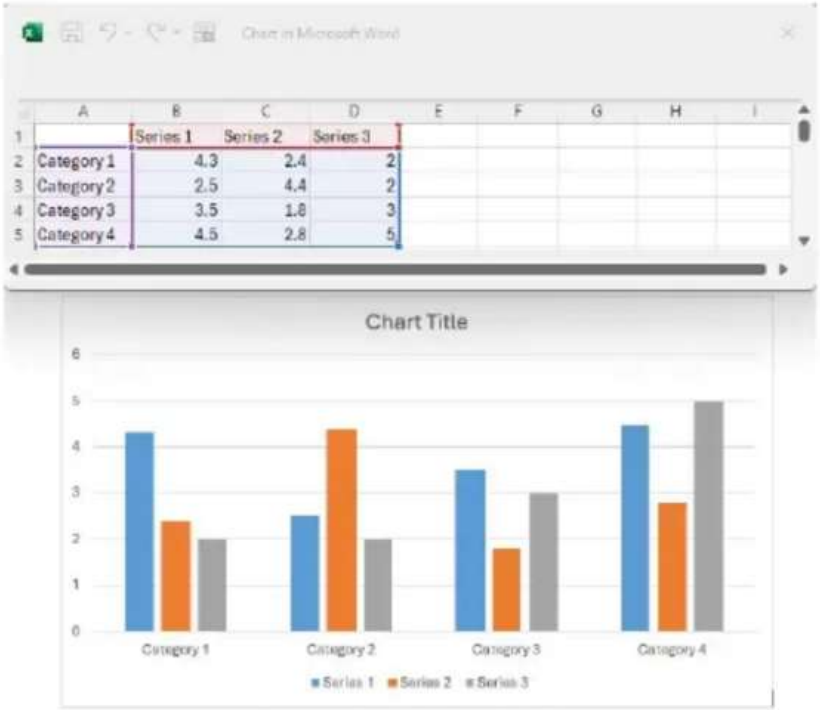


Figure 12:Updating Charts Data

Lab Exercise # 01

1. Create and format a table in MS Word which includes the data of 10 employees. Data includes Employee Name, Department, Performance Score, Feedback Rating, Years of Service and Promotion Status (Yes/No).
   - Sort the table by Performance Scores (highest to lowest), then by Years of Service (longest to shortest) to identify high-performing, long-term employees.
   - Apply formatting options including borders and different table styles to make it visually appealing. Use shading to highlight rows where the Feedback Rating is below 3 (out of 5).
   - Add summary of the overall performance of the employees below the table. Add proper heading, justify the text properly and use font Times New Roman with font size 12.
   - Insert headers and footers and page numbers to the document. The header should include your name, syndicate and the footer should have the company name and your Roll. No.

2. Explain the difference and scope of CPU, GPU, TPU, NPU and ASIC in tabular form, in MS Word. Browse for the features of all, and give proper reference of the websites that you look for, in your lab report.

3. What are the key differences between traditional BIOS and UEFI, and how have these changes improved computer performance and security, give proper reference of the websites that you look for, in your lab report, if any.

Lab 02 Introduction to Microsoft Office Excel

Objective:
Familiarize the students with Microsoft excel and its working.

Software Used:
Microsoft Excel

Microsoft Excel

Microsoft Excel is the most widespread program for creating spreadsheets on the market today. Spreadsheets allow you to organize information in rows and tables (which create cells), with the bonus of automatic mathematics. Spreadsheets have been used for many, many years in business to keep track of expenses and other calculations. Excel will keep track of numbers you place in cells, and if you define cells to refer to each other, any changes made in one cell will be reflected in these referring cells. It sounds a bit complicated, but Excel makes it all a breeze.
Excel works in the form of workbooks. We can create a workbook and then can add as many sheets as possible in the same workbook. We can also rename every sheet according to the data it contains.

1. Opening Microsoft Excel

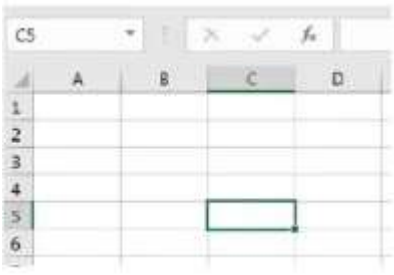
- Open Microsoft Excel from the Start Menu or desktop shortcut.
- Select Blank Workbook to begin with a new sheet.





2. Understanding Cells

- Each Excel sheet consists of rows (numbers) and columns (letters).
- A cell is identified by its column and row (e.g., C5).
- The active cell reference appears in the Name Box (top-left above the sheet).



Note that column C and row 5, and the cell you clicked on are highlighted. “C5” also appears in the top left above the cells. You can move between cells by clicking different cells, or by using the arrow keys.

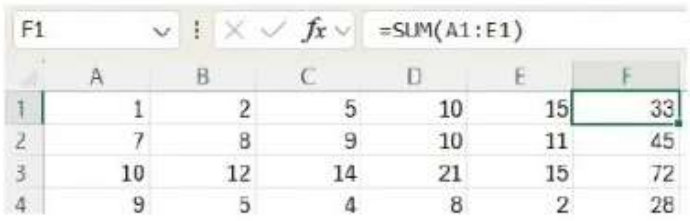
3. Entering Data in Cells

Three things can be entered into each cell: numbers, formulas and text.

- **Numbers:** Type the number (e.g., 123 or 456.7) and press Enter.
- **Formulas**: Begin with = and use operators like +, -, *, /, “and brackets.
  Example: =2+3*4 — Excel follows order of operations (result = 14).
- **Functions:** Built-in formulas for calculations. Common ones:
  =SUM (X: Y) — Adds values between cells X and Y.
  =AVERAGE (X: Y) — Finds mean of values between X and Y.
  =COUNT (X: Y) — Counts filled cells between X and Y.
  =VAR (X: Y) — Calculates sample variance.
  =STDEV (X: Y) — Calculates sample standard deviation.
- **Text**: Type the text and press Enter. If Excel tries to interpret it as a formula/number, start with an apostrophe (’).

Note: Some versions of Excel do not accept the compatibility functions VAR and STDEV. In these cases, you can use =VAR.S(X.Y) and =STDEV.S(X.Y). These commands use identical background formulas. The .S indicates that we are using sample data, rather than population data.

Sum



	A	B	C	D	E	F
1	1	2	5	10	15	33
2	7	8	9	10	11	45
3	10	12	14	21	15	72
4	9	5	4	8	2	28

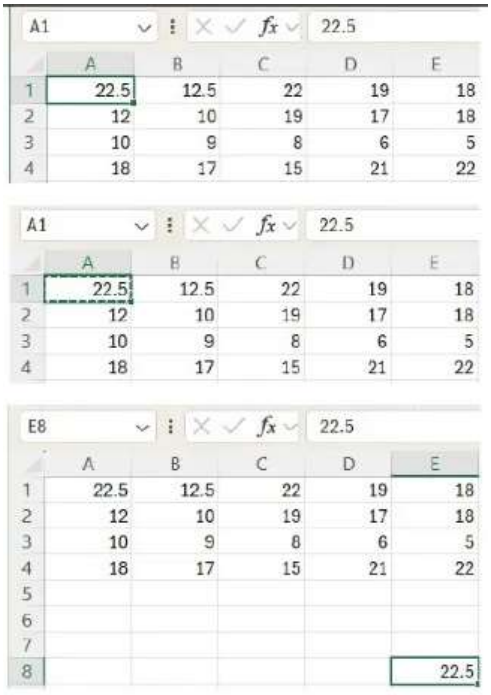
Average



	A	B	C	D	E	F	G
1	3	4	5	7	8	15	
2	8	9	2	4	1	0	5.5
3							

4. Selecting, Copying, Cutting, and Pasting

- Select: Click and drag across cells.
- Copy: Ctrl + C — duplicates data/formulas.
- Cut: Ctrl + X — moves data/formulas.
- Paste: Ctrl + V — inserts copied/cut data into a new location.

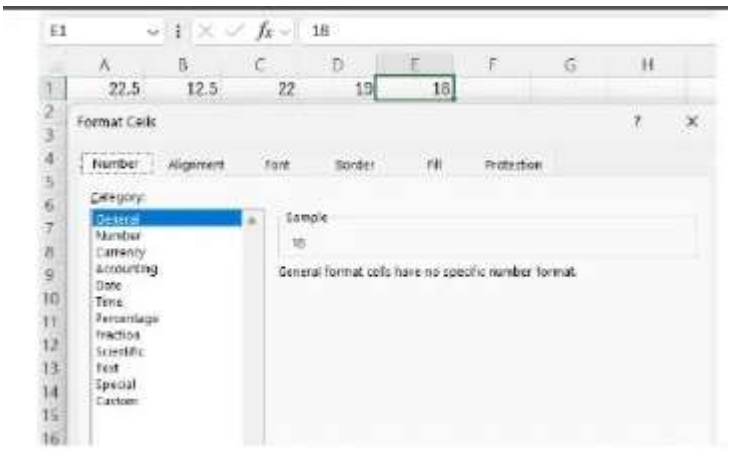


	A	B	C	D	E
1	22.5	12.5	22	19	18
2	12	10	19	17	18
3	10	9	8	6	5
4	18	17	15	21	22
5					
6					
7					
8					22.5

5. Formatting Cells

- Right-click — Format Cells or use the Home tab.
- Options include:

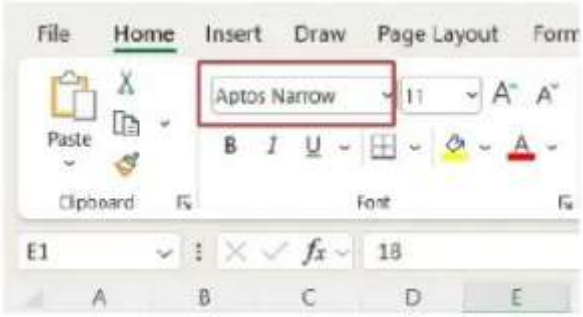
Number — change decimal places or currency format.
Alignment — wrap text inside a cell.
Font — bold, underline, size, style.
Border — draw borders around data for clarity.
Fill — add background color for emphasis.



6. Changing Fonts

- Select the cell(s) you want to format.
- Go to the **Home tab → Font group** (or right-click — Format Cells — Font).
- You can:

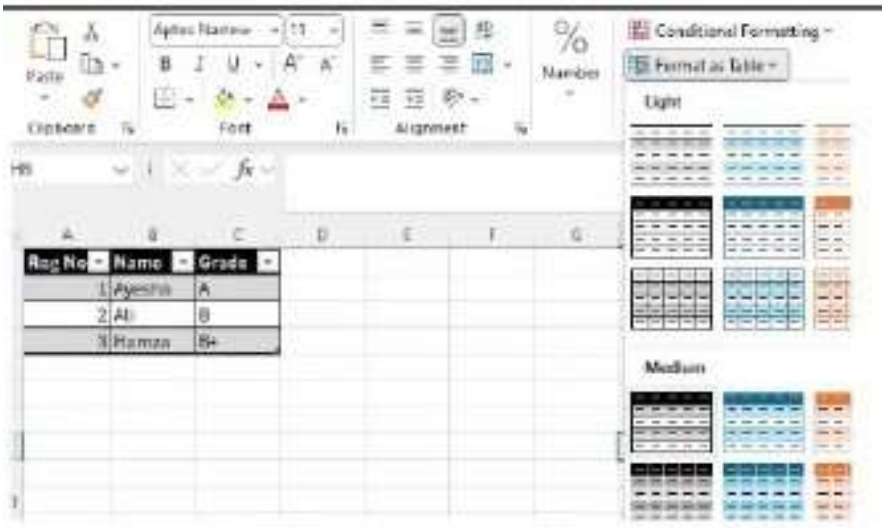
Change the **font type** (e.g., Calibri, Times New Roman).
Adjust the **font size**.
Font — bold, underline, size, style.
Apply **Bold, /italic, or Underline**.
Change the **font color**.



****Note: If you select multiple cells, the formatting will apply to all of them at once.**

7. Format The Table

In Format as tables the selected cells will be converted to table, and our selected color and style formatting will be applied to it.



Lab Exercise # 02

Q1: Create an excel workbook. Insert the following data in it. Format the data so it looks like a table.

Product	Category	Units Sold	Unit Price
Laptop	Electronics	3	800
Smartphone	Electronics	5	600
Headphones	Accessories	10	50
Tablet	Electronics	2	300
Monitor	Electronics	4	200
Keyboard	Accessories	6	30
Mouse	Accessories	8	20

Q2: Modify the above data and insert a column Salesperson.

Product	Category	Salesperson	Units Sold	Unit Price
Laptop	Electronics	John Doe	3	800
Smartphone	Electronics	Jane Smith	5	600
Headphones	Accessories	John Doe	10	50
Tablet	Electronics	Jane Smith	2	300
Monitor	Electronics	Mark Lee	4	200
Keyboard	Accessories	John Doe	6	30
Mouse	Accessories	Jane Smith	8	20

Q3: Create a new column adjacent to the Unit Price with the name “Total amount” for each product with the help of formula.

Q4: Create a new row “Overall Sales” and apply the formula to calculate the overall sales amount.

Lab 03 Data Visualization and Summarization

Objective:

To develop the ability to summarize raw datasets using statistical and analytical functions in Excel, and to represent the summarized information visually through charts and graphs for effective interpretation and decision-making.

Software Used:

Microsoft Excel

Microsoft Excel

Data by itself is often difficult to interpret. Microsoft Excel provides powerful tools for both summarization and visualization, which help in analyzing and presenting information more effectively. Summarization allows us to condense large datasets into meaningful values using formulas and functions such as SUM, AVERAGE, COUNT, MIN, MAX, and STDEV. These summarized results highlight key insights and trends within the data. Visualization, on the other hand, transforms numerical outputs into charts and graphs that make patterns, comparisons, and relationships easier to understand. By using tools such as bar charts, line graphs, pie charts, and scatter plots, data can be communicated in a clear and visually appealing manner. In this lab, we will learn how to combine both summarization and visualization in Excel to turn raw numbers into useful, well-presented information for decision-making and reporting.

1. Opening Microsoft Excel

- Open Microsoft Excel from the Start Menu or desktop shortcut.
- Select Blank Workbook to begin with a new sheet.





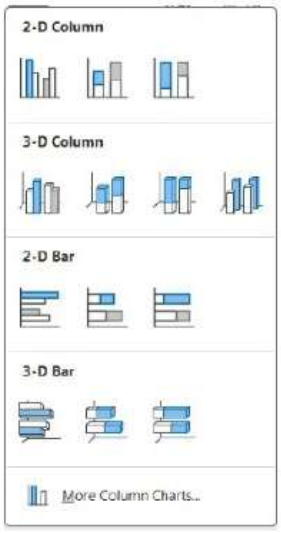
2. Example Dataset

We first need a dataset to work with before creating any visualizations. This lab will use a simple dataset containing sales data for a local electronics store. The dataset contains information on the number of units sold for various product types in 2022 and totals for columns and rows.

	A	B	C	D
1	Month	Television	Laptop	Mobile Phones
2	1/1/2022	145	335	82
3	2/1/2022	145	362	126
4	3/1/2022	105	311	95
5	4/1/2022	171	259	93
6	5/1/2022	178	277	107
7	6/1/2022	167	292	145
8	7/1/2022	200	385	77
9	8/1/2022	181	388	78
10	9/1/2022	152	291	83
11	10/1/2022	143	345	102
12	11/1/2022	114	399	99
13	12/1/2022	109	250	101

3. Creating Basic Charts in Excel

Excel has multiple options for choosing a particular chart type. For example, if you want to create a column or bar chart, you are often presented with various visualization options. For example, there are 2D and 3D versions and normal, stacked, and 100% stacked options. Depending on your requirements, you can choose the visualization type that best suits your needs.

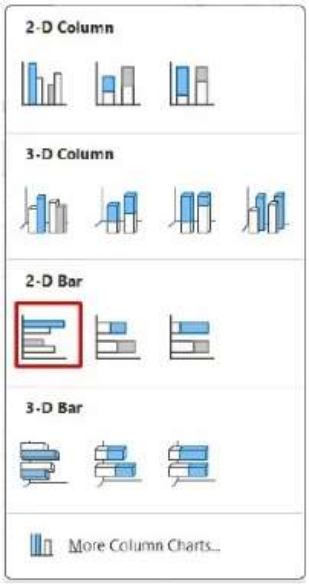


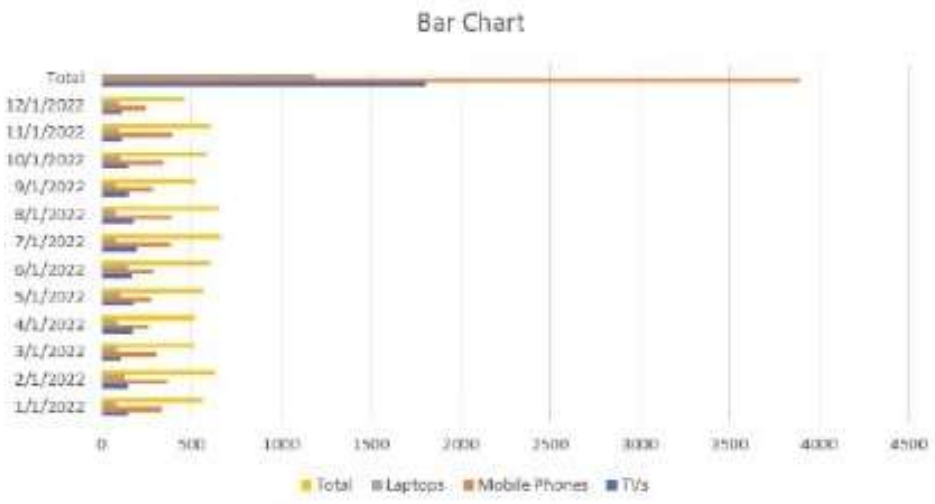
4. Excel bar charts

Bar charts are one of the easiest charts to interpret, enabling the person viewing the chart an easy way to compare categorical data quickly. On a bar chart, the categorical data is on the y-axis, and the values are on the x-axis.

To create a bar chart:

- Select the data range A1:D13
- Click the "Insert" tab in the Excel ribbon
- Click on the column's icon button dropdown, and under the "2-D Bar" category, choose "Clustered Bar''



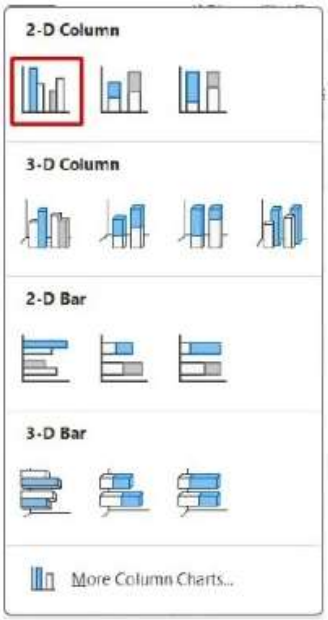


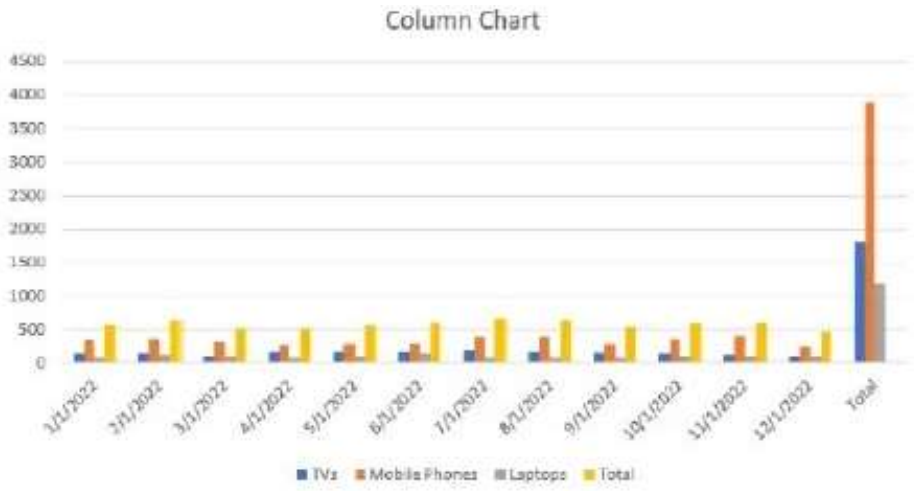
5. Excel column charts

A column chart, also known as a vertical bar chart, helps visualize data where categories are placed on the x-axis and the values on the y-axis. Similar to bar charts, they help visualize data across categories.
To create a column chart in Excel:

- Select the data range A1:D13
- Click the "Insert" tab in the Excel ribbon
- Click on the columns icon dropdown, and under the “2-D Column” category, choose “Clustered Column”

You can now see a column chart that displays the number of units sold for each product category by the month.



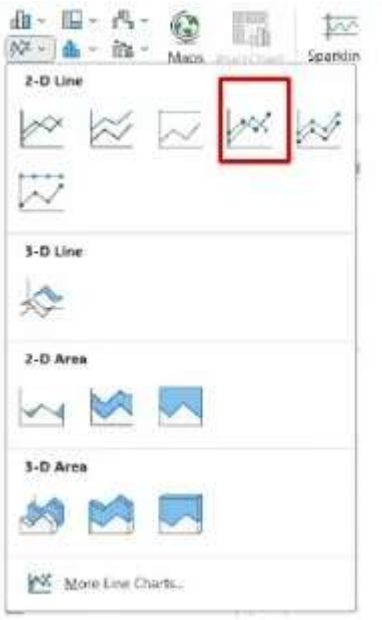


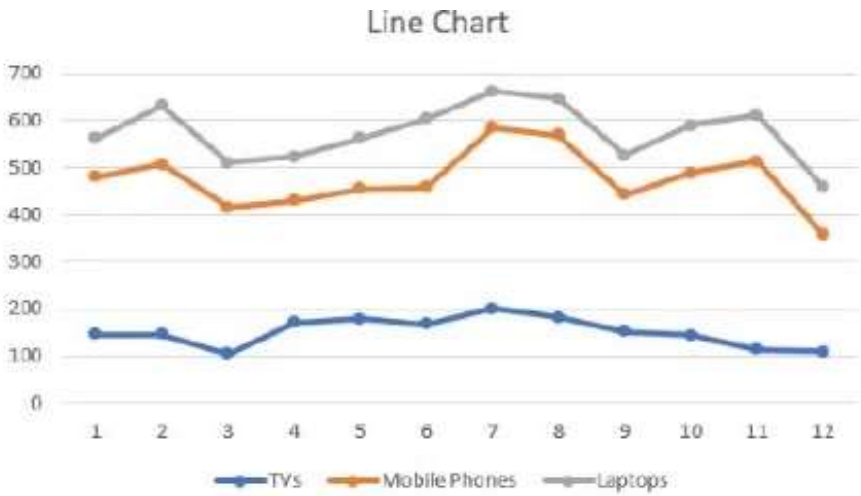
6. Excel line charts

A line chart is the most useful way to capture how a numerical variable changes over time. This is helpful to identify trends in numeric values.
To create a line chart in Excel:

- Select the data range A1:D13
- Click the "Insert" tab in the Excel ribbon
- Click on the line chart dropdown, and under the “2-D Line” category, choose “Line with Markers"

You can now see a line chart displaying units sold each month split by product category. This enables you to compare each product category's performance over time easily.



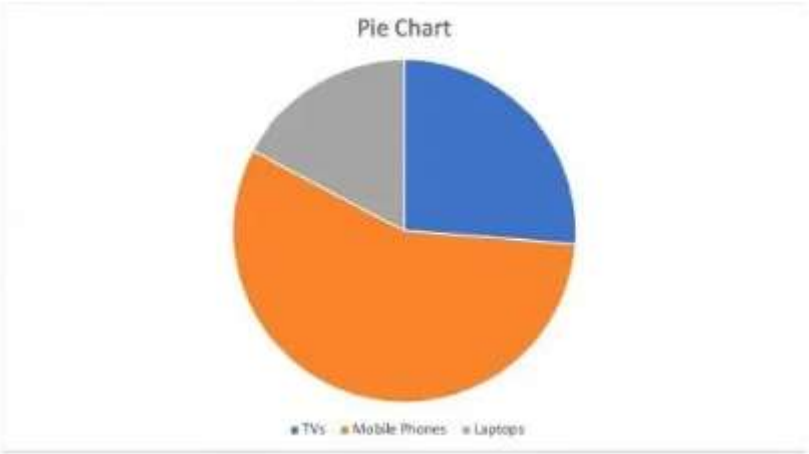


7. Excel pie charts

A pie chart is most commonly used to show the proportions of a whole. It's like visualizing fractions when you were in high school. With this pie chart, we want to compare the total sales between the three categories.

To create a pie chart in Excel:

- First, select the data range B1:D1
- Second, using the command (for Mac) or ctrl (for Windows), select the second date range: B14:D14
- Click the "Insert" tab in the Excel ribbon
- Click on the pie chart dropdown, and under the "2-D Pie" category, choose "Pie" Pie Chart



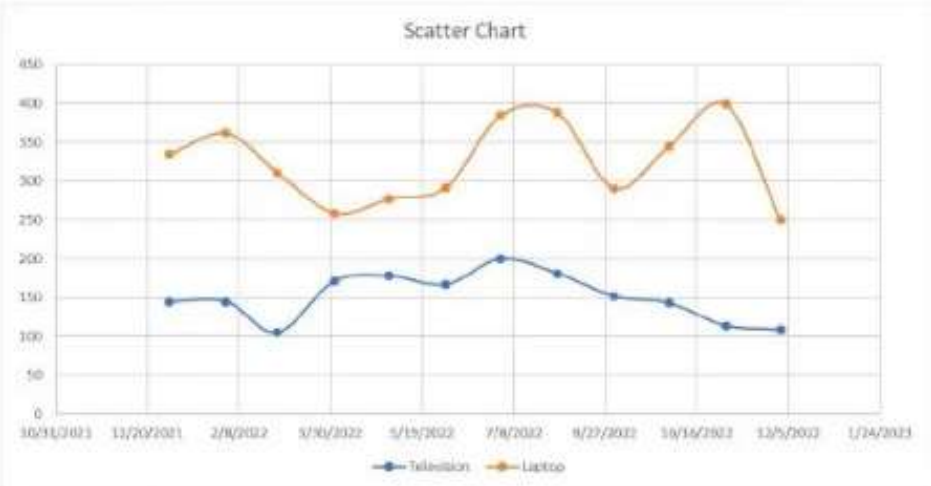
Advanced Excel Visualization Techniques

1. Excel scatter plots

A scatter plot is commonly used to visualize the relationship between two variables. It can be useful for quickly surfacing potential correlations between data points. We'll create a scatter plot to compare the number of TVs and laptops sold.

To create a scatter plot in Excel:

- Select the data range A1:C13
- Click the "Insert" tab in the Excel ribbon
- Click on the scatter plot dropdown, and under the “Scatter” category, choose “Histogram”
- Click "Scatter or Bubble Chart" and choose "Scatter with Smooth Lines and Markers"



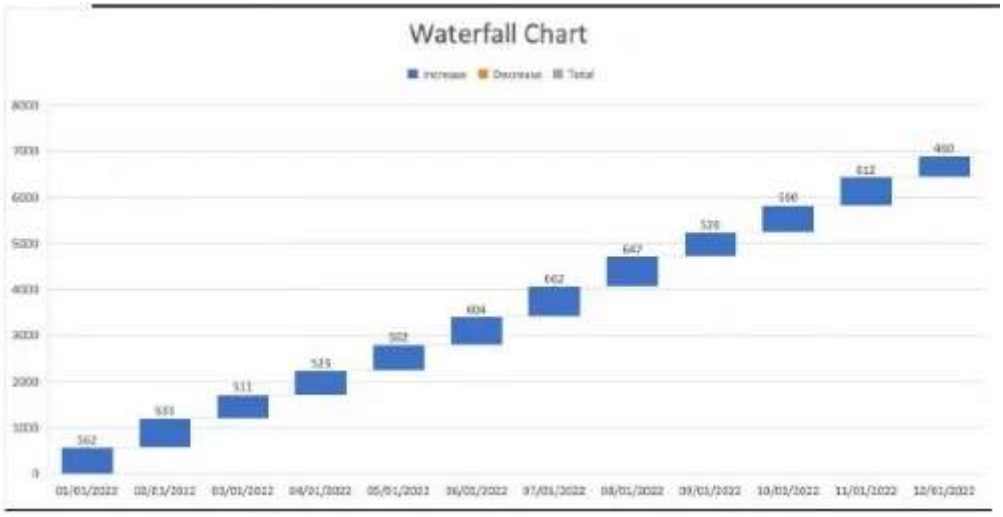
2. Excel waterfall chart

A waterfall chart is a special chart that helps illustrate how positive and negative values can contribute to a total. They can be great for visualizing changes over time. In our example, we'll compare the total sales for each month regardless of the categories.

To create a waterfall chart in Excel:

- First, select the data range A2:A13.
- Second, using the command (for Mac) or ctrl (for Windows), select the second data range E2:E13.
- Click on the waterfall chart dropdown, and under the “Waterfall” category, choose “Waterfall”.

We'll only see positive values in our example because the Total column only contains positive values, but this chart can be great for comparing financial data and changes over time.



3. Chart elements

Chart elements include titles, legends, data labels, gridlines, and axes. You can add, remove, or modify these elements as needed.
Click on the chart to select it.

- In the Excel ribbon, click the "Chart Design" tab
- Click the "Add Chart Element" dropdown to access the available chart elements



4. Chart title

- To add a chart title, click "Add Chart Element" > "Chart Title" and choose one of the available options (e.g., "Above Chart" or "Centered Overlay"). You can then click the title placeholder and type your desired title
- To remove a chart title, right-click on the title and select “Delete”

For this lab, rename the chart “Electronic Store Sales 2022”

5. Legend

- To modify the chart legend, click "Add Chart Element" > "Legend" and choose a position for the legend (e.g., "Right" or "Top")
- To remove the legend, click "Add Chart Element" > "Legend" > "None"

For this tutorial, move the legend to the top of the chart.

6. Data labels

- To add data labels, click "Add Chart Element" > "Data Labels" and choose one of the available options (e.g., "Center" or "Above"). This will display the data values directly on the chart
- To remove data labels, click “Add Chart Element” > “Data Labels” > “None”
- To remove data labels from individual categories, right-click on the data label of the chosen category; this will highlight all relevant data labels. You can then click ‘’Delete’’, For this lab, add data labels for all categories.

Your chart should now look like this:



Lab Exercise # 03

Q1: Create a 2-D Column Chart of the electronics dataset to show monthly sales of all three products.

Q2: Make a 3-D Bar Chart of the same dataset and compare it with the column chart. Which looks clearer?

Q3: Draw a Line Chart for TV sales (Jan—Dec 2022) to observe the trend.

Q4: Prepare a Pie Chart to show the share of TVs, Mobiles, and Laptops in total yearly sales.

Q5: Insert a Scatter Plot to compare TV sales with Laptop sales.

Q6: Format any one chart by adding: Chart Title, Axis Titles, Legend, and Data Labels.

Lab 04 Microsoft Excel: Data Analytics

Objective:

The objective of this lab is to enable students to perform basic data analytics using Excel. Students will learn how to prepare and clean datasets, apply sorting, filtering, and conditional formatting to identify patterns, and use built-in functions such as COUNTIF, SUMIF, AVERAGEIF, and VLOOKUP for analysis. The lab also aims to develop skills in summarizing data through PivotTables and visualizing insights using charts, helping students make data- driven decisions effectively.

Software Used:

- Microsoft Excel

Microsoft Excel

Excel is one of the most powerful tools for data analysis, allowing you to process, manipulate, and visualize large datasets efficiently. Whether you're analyzing sales figures, financial reports, or any other type of data, knowing how to perform data analysis in Excel can help you make informed decisions quickly. In this guide, we will walk you through the essential steps to analyze data in Excel, including using built-in tools like pivot tables, charts, and formulas to extract meaningful insights. By the end of this article, you'll have a solid understanding of how to use Excel for data analysis and improve your decision-making process.

Opening Microsoft Excel

- Open Microsoft Excel from the Start Menu or desktop shortcut.
- Select Blank Workbook to begin with a new sheet.





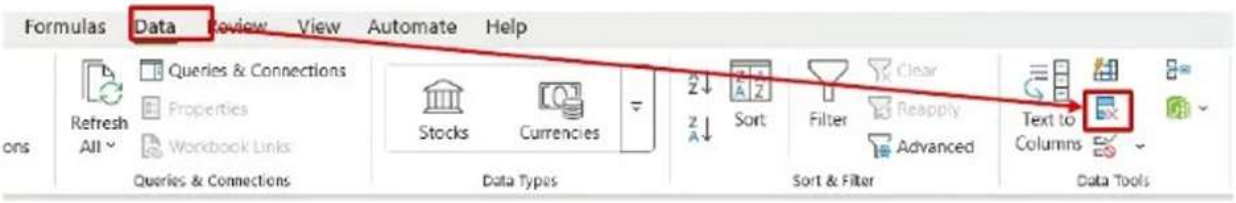
Preparing Data for Analysis

Data cleaning becomes an intuitive process with Excel's capabilities, allowing users to identify and rectify issues like missing values and duplicates. PivotTables, a hallmark feature, empowers users to swiftly summarize and explore large datasets, providing dynamic insights through customizable cross-tabulations, making Data Analysis Excel an essential skill for professionals.

How to Clean Data in Excel

1. Remove Duplicates

- Select the row or a column
- Go to the Data tab on the Ribbon.
- Click Remove Duplicates (in the Data Tools group).
- A dialog box will appear - select the columns you want Excel to check for duplicates.
- Click OK - Excel will remove duplicate entries and display a message showing how many were removed.



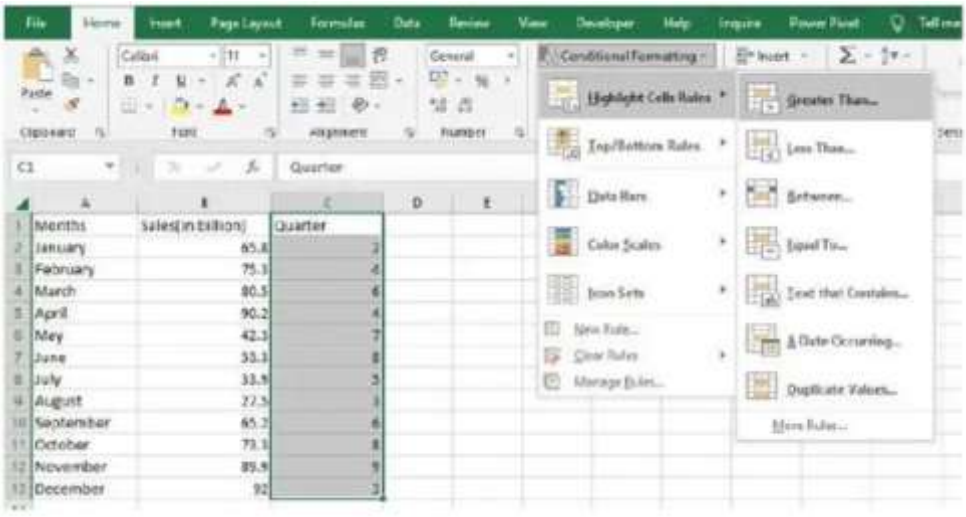
10	12	10
11	13	10
12	21	9
5	11	16
19	12	6
	5	7

2. Use TRIM Function (remove extra spaces)

- Suppose your data is in column A (e.g., names with extra spaces).
- In an empty column (say column L), type: =TRIM(K1)
- Press Enter → this will remove all extra spaces except single spaces between words.
- Copy the formula down for all rows.
- Copy the cleaned column (B), then Paste as Values back into column A (original data).

Go to Home → Conditional Formatting

Select any column from the table. Here we are going to select a Quarter column. After that go to the home tab on the top of the ribbon and then in the styles group select conditional formatting and then in the highlight cells rule select Greater than an option.



Then a greater than dialog box appears. Here first write the quarter value and then select the color.



	A	B	C	D
1	Months	Sales(in billion)	Quarter	
2	January	65.8	2	
3	February	75.3	4	
4	March	80.5	6	
5	April	90.2	4	
6	May	42.3	7	
7	June	53.3	8	
8	July	33.9	5	
9	August	27.5	3	
10	September	65.2	6	
11	October	73.3	8	
12	November	89.9	9	
13	December	92	3	
14				
15				

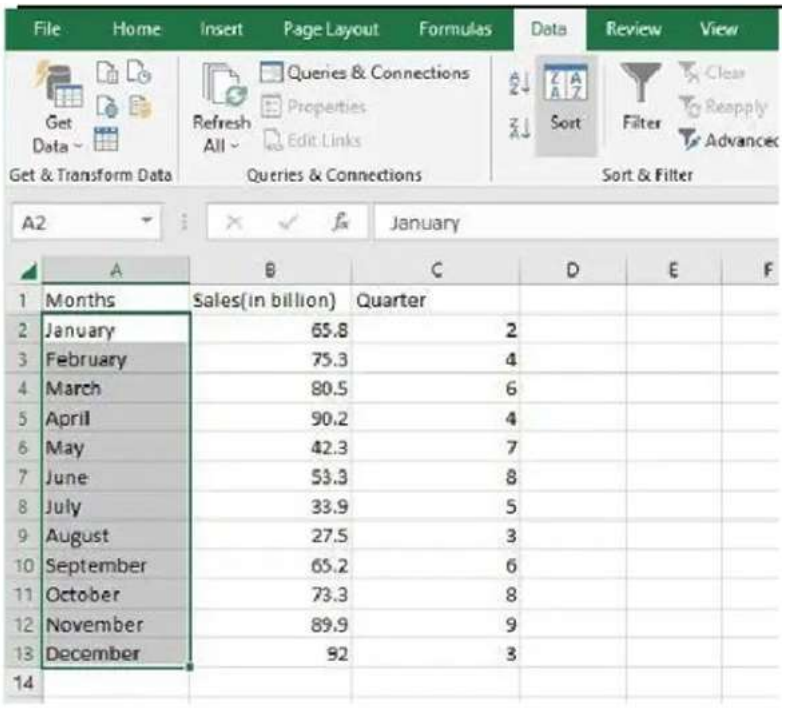
4. Sorting Data By Column

Data analysis Excel requires sorting the data. A list of names may be arranged alphabetically, a list of sales numbers can be arranged from highest to lowest, or rows can be sorted by colors or icons. Sorting data makes it easier to immediately view and comprehend your data, organize and locate the facts you need, and ultimately help you make better decisions. Both columns and rows can be used to sort. You'll utilize column sorts for the majority of your sorting. By text, numbers, dates, and times, a custom list, format, including cell color, font color, or icon set, you may sort data in one or more columns.

- **Single Column**: Sort data alphabetically or numerically using Data > Sort.
- **Multiple Columns:** Perform multi-level sorting by adding criteria in the sort dialog box.

Select Data > Data Tab> Sort

Select any column from the table. Here we are going to select a Month's column. After that go to the data tab on the top of the ribbon and then in the sort and filters group select sort.



	A	B	C	D	E	F
1	Months	Sales(in billion)	Quarter			
2	January	65.8	2			
3	February	75.3	4			
4	March	80.5	6			
5	April	90.2	4			
6	May	42.3	7			
7	June	53.3	8			
8	July	33.9	5			
9	August	27.5	3			
10	September	65.2	6			
11	October	73.3	8			
12	November	89.9	9			
13	December	92	3			
14						

Select the Order

Then a sort dialog box appears. Here first select the column, and then select sort on, and then Order. After that, click OK.



Preview Results

Now as you can see the months column is now arranged alphabetically.

	A	B	C	D
1	Months	Sales(in billion)	Quarter	
2	April	65.8	2	
3	August	75.3	4	
4	December	80.5	6	
5	February	90.2	4	
6	January	42.3	7	
7	July	53.3	8	
8	June	33.9	5	
9	March	27.5	3	
10	May	65.2	6	
11	November	73.3	8	
12	October	89.9	9	
13	September	92	3	
14				
15				

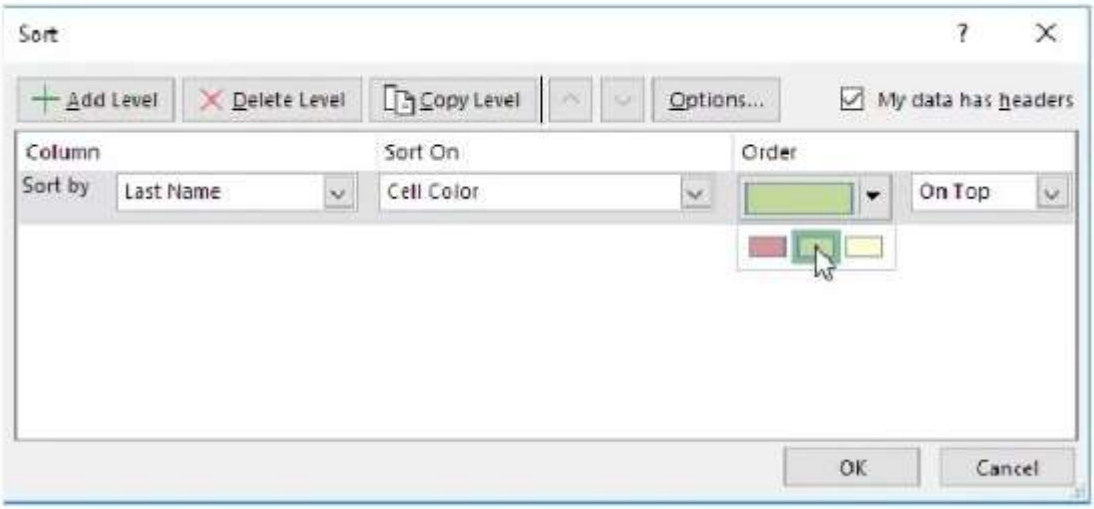
5. Sort by Color in Excel

This example teaches you how to sort data by color in Excel.

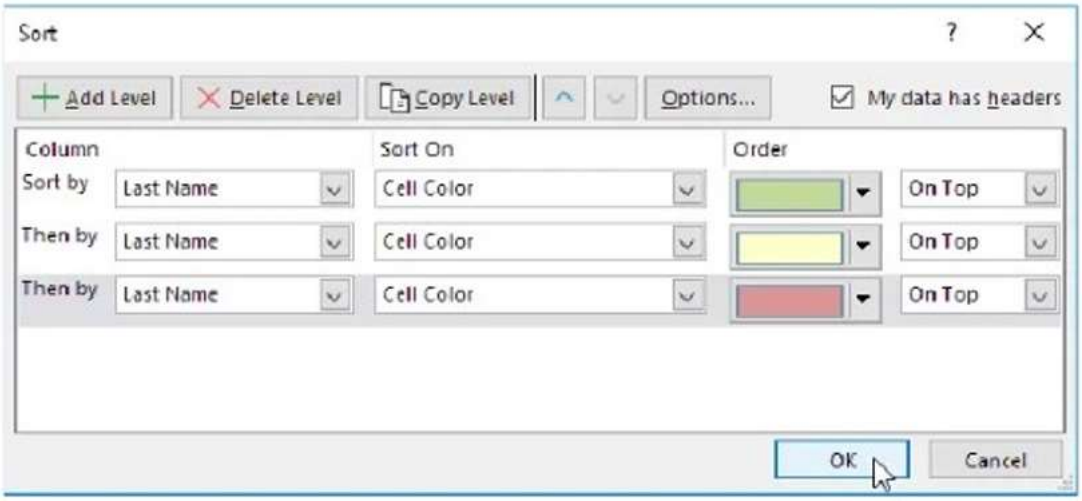
	A	B	C	D	E
1	Last Name	Sales	Country	Quarter	
2	Smith	$16,753.00	UK	Qtr 3	
3	Johnson	$14,808.00	USA	Qtr 4	
4	Williams	$10,644.00	UK	Qtr 2	
5	Jones	$1,390.00	USA	Qtr 3	
6	Brown	$4,865.00	USA	Qtr 4	
7	Williams	$12,438.00	UK	Qtr 1	
8	Johnson	$9,339.00	UK	Qtr 2	
9	Smith	$18,919.00	USA	Qtr 3	
10	Jones	$9,213.00	USA	Qtr 4	
11	Jones	$7,433.00	UK	Qtr 1	
12	Brown	$3,255.00	USA	Qtr 2	
13	Williams	$14,867.00	USA	Qtr 3	
14	Williams	$19,302.00	UK	Qtr 4	
15	Smith	$9,698.00	USA	Qtr 1	
16					

1. Click any single cell inside a data set.
2. On the Data tab, in the Sort & Filter group, click Sort.

- The Sort dialog box appears.
- Sort by Last Name (or any other column), sort on Cell Color (you can also sort on Font Color and Cell Icon), and select the green color for the first level.



- Click ‘Copy Level’ two times and select the other colors then click Ok a shown below



- **Result**

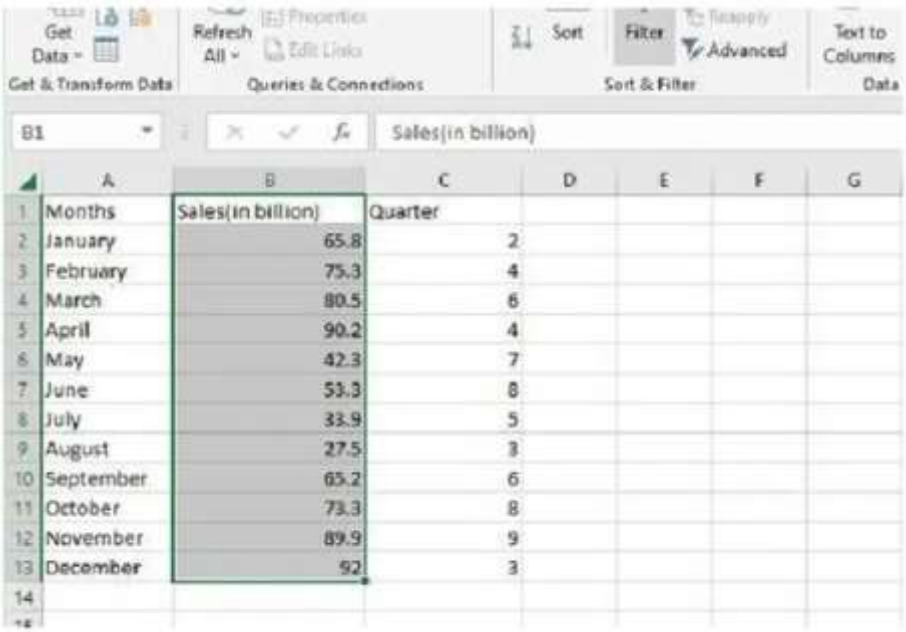
3	Brown	$4,865.00	USA	Qtr 4
4	Smith	$18,919.00	USA	Qtr 3
5	Jones	$9,213.00	USA	Qtr 4
6	Williams	$14,867.00	USA	Qtr 3
7	Williams	$19,302.00	UK	Qtr 4
8	Smith	$16,753.00	UK	Qtr 3
9	Williams	$12,438.00	UK	Qtr 1
10	Jones	$7,433.00	UK	Qtr 1
11	Williams	$10,644.00	UK	Qtr 2
12	Jones	$1,390.00	USA	Qtr 3
13	Johnson	$9,339.00	UK	Qtr 2
14	Brown	$3,255.00	USA	Qtr 2
15	Smith	$9,698.00	USA	Qtr 1
16				

6. Filtering Data

You may use filtering to pull information from a given Range or table that satisfies the specified criteria in data analysis excel. This is a fast method of just showing the data you require. Data in a Range, table, or PivotTable may be filtered. You may use Selected Values to filter data. You may adjust your filtering options in the Custom AutoFilter dialogue box that displays when you click a Filter option or the Custom Filter link that is located at the end of the list of Filter options.

Select your dataset and go to Data > Filter

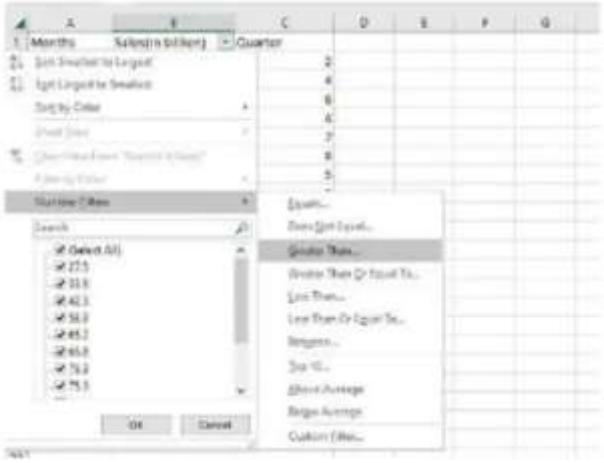
Select any column from the table. Here we are going to select a Sales column. After that go to the data tab on the top of the ribbon and then in the sort and filters group select filter.



	A	B	C	D	E	F	G
1	Months	Sales(in billion)	Quarter				
2	January	65.8	2				
3	February	75.3	4				
4	March	80.5	6				
5	April	90.2	4				
6	May	42.3	7				
7	June	53.3	8				
8	July	33.9	5				
9	August	27.5	3				
10	September	65.2	6				
11	October	73.3	8				
12	November	89.9	9				
13	December	92	3				
14							

Select the Filter Option

The values in the sales column are then shown in a drop-down box. Here we are going to select a number of filters and then greater than.



Then a custom auto filler dialog box appears. Here we are going to apply sales greater than 70 and then click OK.



Preview Results

Now as you can see only the rows greater than 70 are shown.

	A	B	C	D
1	Months	Sales(in billion)	Quarter	
3	February	75.3	4	
4	March	80.5	6	
5	April	90.2	4	
11	October	73.3	8	
12	November	89.9	9	
13	December	92	3	
14				
15				
16				

Excel Functions to Simplify Data Analysis

Excel's built-in functions allow for quick calculations and summaries:

1. LEN

=LEN quickly returns the character count in a given cell. The =LEN formula may be used to calculate the number of characters needed in a cell to distinguish between two different kinds of product Stock Keeping Units, as seen in the example above. When trying to discern between different Unique Identifiers, which might occasionally be lengthy and out of order, LEN is very crucial.

=LEN(Select Cell)

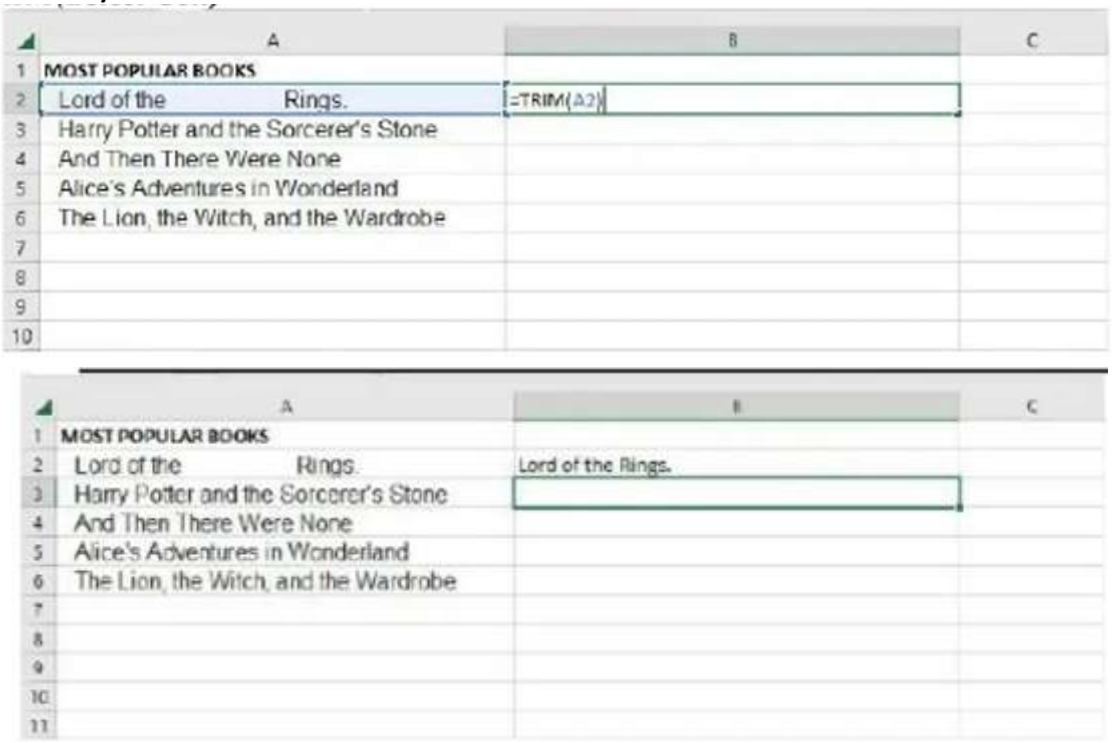
	A	B	C
1	MOST POPULAR BOOKS		
2	Lord of the Rings.	=LEN(A2)	
3	Harry Potter and the Sorcerer's Stone		
4	And Then There Were None		
5	Alice's Adventures in Wonderland		
6	The Lion, the Witch, and the Wardrobe		
7			
8			
9			

	A	B	C
1	MOST POPULAR BOOKS		
2	Lord of the Rings.	19	
3	Harry Potter and the Sorcerer's Stone		
4	And Then There Were None		
5	Alice's Adventures in Wonderland		
6	The Lion, the Witch, and the Wardrobe		
7			
8			
9			
10			

2. TRIM

=TRIM function will remove all spaces from a cell, with the exception of single spaces between words. The most frequent application of this function is to get rid of trailing spaces. When content is copied verbatim from another source or when users insert spaces at the end of the text, this is normal.

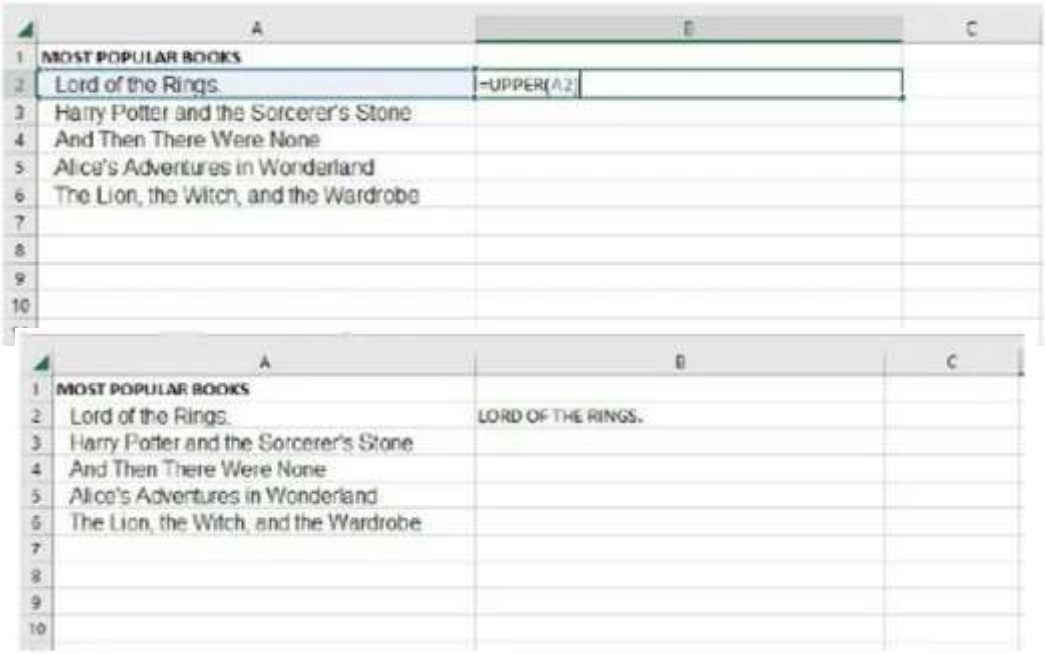
=TRIM (Select Cell)



3. UPPER

The Excel Text function "UPPER Function" will change the text to all capital letters (UPPERCASE). As a result, the function changes all of the characters in a text string input to upper case.

=UPPER (Text)



4. COUNTIF

Excel has a built-in function called COUNTIF that counts the given cells. The COUNTIF function can be used in both straightforward and sophisticated applications in data analysis excel. The fundamental application of counting particular numbers and words is covered in this.

=COUNTIF (range, criteria)

Range: The size of the cell range to count.

Criteria: The standards by which cells are selected for counting. To count the number of regions of each type, apply the COUNTIF function to the range B2:B20.

	A	B	C	D	E	F	G	H
1	Country	Region	Days					
2	Russia	East	1					
3	Russia	East	4					
4	Switzerland	West	1					
5	Switzerland	West	2			East		
6	Switzerland	West	1			West		
7	New Zealand	East	1			North		
8	New Zealand	East	1			South		
9	Canada	North	10			Northeast		
10	Canada	North	3					
11	Canada	North	7					
12	Canada	North	8					
13	UK	West	1					
14	UK	West	1					
15	Greenland	North	1					
16	Peru	South	1					
17	Peru	South	2					
18	Nagaland	Northeast	1					
19	Brazil	South	1					
20	Kohima	Northeast	1					
21								

Next, use the COUNTIF function on the range F5:F9 to count the different types of regions.

	A	B	C	D	E	F	G	H
1	Country	Region	Days					
2	Russia	East	1					
3	Russia	East	4					
4	Switzerland	West	1					
5	Switzerland	West	2			East	=COUNTIF(B2:B20,F5)	
6	Switzerland	West	1			West		
7	New Zealand	East	1			North		
8	New Zealand	East	1			South		
9	Canada	North	10			Northeast		
10	Canada	North	3					
11	Canada	North	7					
12	Canada	North	8					
13	UK	West	1					
14	UK	West	1					
15	Greenland	North	1					
16	Peru	South	1					
17	Peru	South	2					
18	Nagaland	Northeast	1					
19	Brazil	South	1					
20	Kohima	Northeast	1					
21								

As you can see, the COUNTIF function has correctly enumerated the number of regions, such as the 4 East Regions, as expected.

	A	B	C	D	E	F	G	H
1	Country	Region	Days					
2	Russia	East	1					
3	Russia	East	4					
4	Switzerland	West	1					
5	Switzerland	West	2			East	4	
6	Switzerland	West	1			West		
7	New Zealand	East	1			North		
8	New Zealand	East	1			South		
9	Canada	North	10			Northeast		
10	Canada	North	3					
11	Canada	North	7					

To calculate the average speed of vehicles, apply the AVERAGEIF function on the range B2:B10.

	A	B	C	D	E	F	G	H	I	J
1	Country	Vehicle	Speed							
2	Russia	Car	60							
3	Nagaland	Bus	70							
4	Switzerland	Truck	75				Type	Average Speed		
5	UK	Car	59				Car	=AVERAGEIF(B2:B10,G5,C2:C10)		
6	GreenLand	Car	68				Bus			
7	New Zealand	Bus	75				Truck			
8	Brazil	Truck	80							
9	Kohima	Truck	82							
10	Canada	Bus	76							
11										

	A	B	C	D	E	F	G	H	I
1	Country	Vehicle	Speed						
2	Russia	Car	60						
3	Nagaland	Bus	70						
4	Switzerland	Truck	75				Type	Average Speed	
5	UK	Car	59				Car	62.33333333	
6	GreenLand	Car	68				Bus		
7	New Zealand	Bus	75				Truck		
8	Brazil	Truck	80						
9	Kohima	Truck	82						
10	Canada	Bus	76						

5. SUMIF

A built-in Excel function called SUMIF determines if a condition is true or false before adding the values in a range.

=**SUMIF(range, criteria, [sum_range])**

Range: The size of the cell range to count.

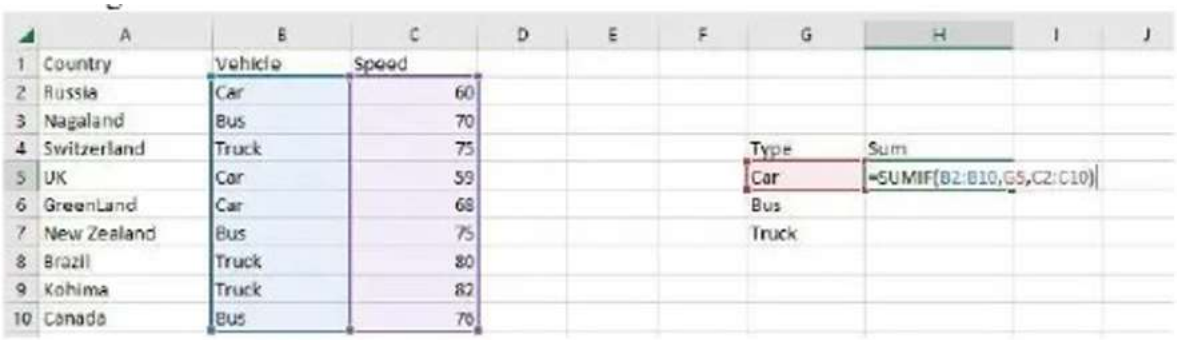
Criteria: The standards by which cells are selected for counting.

Sum Range: The range that the function uses to calculate the total is known as the sum range.

To calculate the sum of the vehicle's speed, apply the SUMIF function on the range B2:B10.

	A	B	C	D	E	F	G	H	I
1	Country	Vehicle	Speed						
2	Russia	Car	60						
3	Nagaland	Bus	70						
4	Switzerland	Truck	75				Type	Sum	
5	UK	Car	59				Car		
6	GreenLand	Car	68				Bus		
7	New Zealand	Bus	75				Truck		
8	Brazil	Truck	80						
9	Kohima	Truck	82						
10	Canada	Bus	76						

Next, use the SUMIF function on the range H4:H7 to find the sum of the vehicle speeds listed in that range.



	A	B	C	D	E	F	G	H	I	J
1	Country	Vehicle	Speed							
2	Russia	Car	60							
3	Nagaland	Bus	70							
4	Switzerland	Truck	75				Type	Sum		
5	UK	Car	59				Car	=SUMIF(B2:B10,G5,C2:C10)		
6	GreenLand	Car	68				Bus			
7	New Zealand	Bus	75				Truck			
8	Brazil	Truck	80							
9	Kohima	Truck	82							
10	Canada	Bus	76							

As you can see, the SUMIF function has correctly calculated the total speed, such as the 187 sum for cars.



	A	B	C	D	E	F	G	H	I
1	Country	Vehicle	Speed						
2	Russia	Car	60						
3	Nagaland	Bus	70						
4	Switzerland	Truck	75				Type	Sum	
5	UK	Car	59				Car	187	
6	GreenLand	Car	68				Bus		
7	New Zealand	Bus	75				Truck		
8	Brazil	Truck	80						
9	Kohima	Truck	82						
10	Canada	Bus	76						

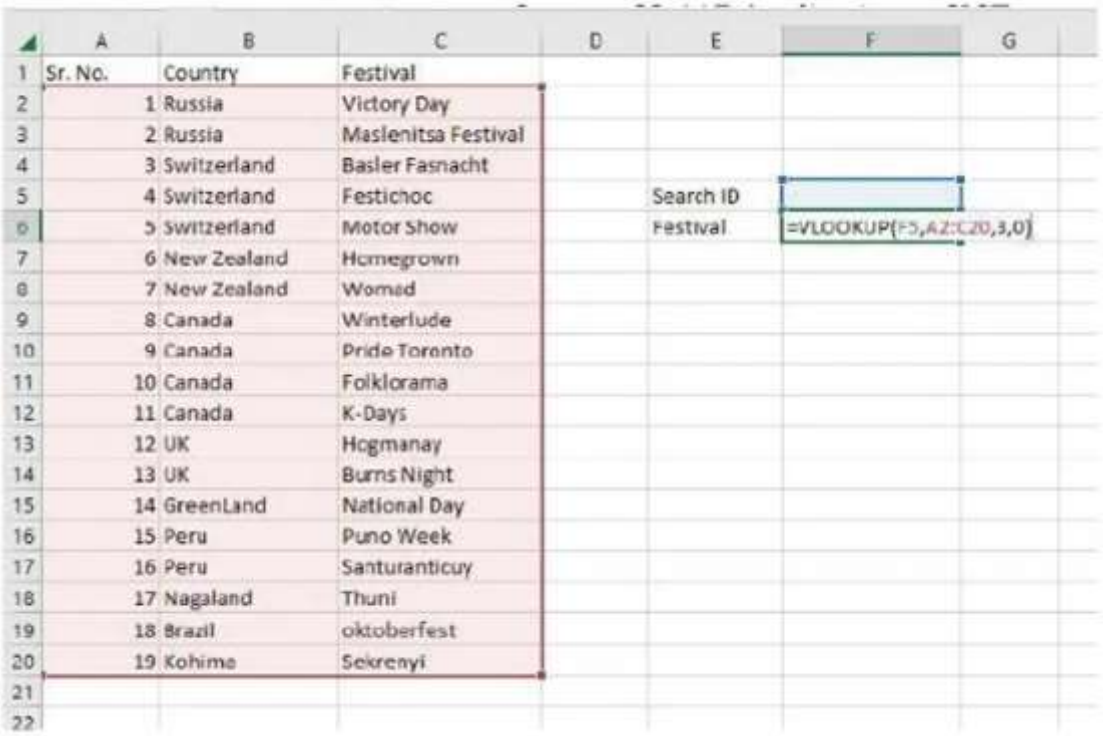
***Range_lookup**: FALSE if text (0), TRUE if numbers (1).*

To locate the festival names based on their search ID, use the VLOOKUP function. The festival names will be determined by their corresponding search ID.



	A	B	C	D	E	F	G
1	Sr. No.	Country	Festival				
2	1	Russia	Victory Day				
3	2	Russia	Maslanitsa Festival				
4	3	Switzerland	Basler Fasnacht				
5	4	Switzerland	Festichoc		Search ID		
6	5	Switzerland	Motor Show		Festival		
7	6	New Zealand	Homegrown				
8	7	New Zealand	Womad				
9	8	Canada	Winterlude				
10	9	Canada	Pride Toronto				
11	10	Canada	Folklorama				
12	11	Canada	K-Days				
13	12	UK	Hogmanay				
14	13	UK	Burns Night				
15	14	GreenLand	National Day				
16	15	Peru	Puno Week				
17	16	Peru	Santuranticuy				
18	17	Nagaland	Thuni				
19	18	Brazil	oktoberfest				
20	19	Kohima	Sekrenyi				
21							

In cell F5, enter the search query (lookup value). The table array range is A2:C20, with the col index number set to 3 (since the information is in the third column from the left). Set the range lookup to 0 (False) to ensure an exact match.



	A	B	C	D	E	F	G
1	Sr. No.	Country	Festival				
2	1	Russia	Victory Day				
3	2	Russia	Maslenitsa Festival				
4	3	Switzerland	Basler Fasnacht				
5	4	Switzerland	Festichoc		Search ID		
6	5	Switzerland	Motor Show		Festival	=VLOOKUP(F5,A2:C20,3,0)	
7	6	New Zealand	Homegrown				
8	7	New Zealand	Womad				
9	8	Canada	Winterlude				
10	9	Canada	Pride Toronto				
11	10	Canada	Folklorama				
12	11	Canada	K-Days				
13	12	UK	Hogmanay				
14	13	UK	Burns Night				
15	14	GreenLand	National Day				
16	15	Peru	Puno Week				
17	16	Peru	Santuranticuy				
18	17	Nagaland	Thuni				
19	18	Brazil	oktoberfest				
20	19	Kohima	Sekrenyi				
21							
22							

If the lookup value in F5 does not exist in the table, the VLOOKUP function will return a #N/A value, indicating that no match was found.

	A	B	C	D	E	F	G
1	Sr. No.	Country	Festival				
2	1	Russia	Victory Day				
3	2	Russia	Maslenitsa Festival				
4	3	Switzerland	Basler Fasnacht				
5	4	Switzerland	Festichoc		Search ID		
6	5	Switzerland	Motor Show		Festival	#N/A	
7	6	New Zealand	Homegrown				
8	7	New Zealand	Womad				
9	8	Canada	Winterlude				
10	9	Canada	Pride Toronto				
11	10	Canada	Folklorama				
12	11	Canada	K-Days				
13	12	UK	Hogmanay				
14	13	UK	Burns Night				
15	14	GreenLand	National Day				
16	15	Peru	Puno Week				
17	16	Peru	Santuranticuy				
18	17	Nagaland	Thuni				
19	18	Brazil	oktoberfest				
20	19	Kohima	Sekrenyi				
21							



	A	B	C	D	E	F	G
1	Sr. No.	Country	Festival				
2	1	Russia	Victory Day				
3	2	Russia	Maslenitsa Festival				
4	3	Switzerland	Basler Fasnacht				
5	4	Switzerland	Festichoc		Search ID	6	
6	5	Switzerland	Motor Show		Festival	Homegrown	
7	6	New Zealand	Homegrown				
8	7	New Zealand	Womad				
9	8	Canada	Winterlude				
10	9	Canada	Pride Toronto				
11	10	Canada	Folklorama				
12	11	Canada	K-Days				
13	12	UK	Hogmanay				
14	13	UK	Burns Night				
15	14	GreenLand	National Day				
16	15	Peru	Puno Week				
17	16	Peru	Santuranticuy				
18	17	Nagaland	Thuni				
19	18	Brazil	oktoberfest				
20	19	Kohima	Sekrenyi				
21							

Lab Task 04

Dataset: Sales Records

	A	B	C	D	E	F
1	**ID**	**Name**	**Region**	**Product**	**Sales**	**Month**
2	101	John Smith	East	Laptop	450	January
3	102	Anna Jones	West	Mobile	720	February
4	103	Peter Brown	East	Laptop	510	March
5	104	John Smith	East	Laptop	450	March
6	105	Sara Khan	North	Tablet	380	April
7	106	Anna Jones	West	Mobile	720	May
8	107	David Lee	South	Laptop	650	June
9	108	Peter Brown	East	Mobile	820	July
10	109	Sara Khan	North	Tablet	380	August
11	110	John Smith	East	Laptop	450	September

Q1: Data Cleaning

- Remove the duplicate records from the dataset (john Smith and Anna Jones are repeated).
- Use the TRIM function to clean spaces in the Name column.
- Convert all names into UPPERCASE using the UPPER function.

Q2: Sorting and Filtering

- Sort the dataset by Sales (highest to lowest).
- Sort the dataset alphabetically by Region.
- Apply a Filter to display only sales greater than 500.

Q3: Conditional Formatting

- Highlight sales values greater than 600 in green.
- Highlight sales values less than 400 in red.

Q4: Excel Functions

- Use LEN to count the number of characters in the Name field for the first three rows.
- Use COUNTIF to count how many sales were made in the East region.
- Use SUMIF to calculate the total sales for Mobile products.
- Use AVERAGEIF to find the average sales for Laptops.

Q5: VLOOKUP

- Create a lookup table where entering an ID returns the corresponding Region.
  Example: Input = 103 → Output = East.

** NOTE: Use the above given data set under the lab task heading (Sales Record)*

Lab 05 Microsoft Office (Apps and Cloud)

Objective

The objective of this lab is to enable students to use Microsoft Office Online (Apps and Cloud) effectively. Students will learn how to log in to the Office Online platform using their Microsoft account; access and use online applications such as Word, PowerPoint, Excel, and Outlook; understand the benefits of using Office in the cloud, including real-time collaboration, auto-save, and cross-device accessibility; and work with documents directly in the browser without needing the desktop version of Office.

Software Used

- Microsoft Office Online: www.office.com

Microsoft Office Online

Microsoft Office Online is a cloud-based platform that provides access to popular Office applications such as Word, Excel, PowerPoint, and Outlook through a web browser. Instead of installing software on your computer, you can create, edit, and share documents directly online.
Key advantages of Office Online include:

- Cross-device support: Works on any device with internet access.
- Auto-save: Changes are saved automatically to OneDrive.
- Collaboration: Multiple users can work on the same file in real time.
- Collaboration: Share documents and edit simultaneously.

Logging into Microsoft Office Online

1. Open a web browser such as Chrome, Edge, or Firefox.
2. Go to www.office.com.
3. Click Sign in.
4. Enter your Microsoft account email, such as an Outlook, Hotmail, or institutional email.
5. Type your password and click Next.
6. Once logged in, the Office Online dashboard will appear with app icons for Word, Excel, PowerPoint, Outlook, and more.

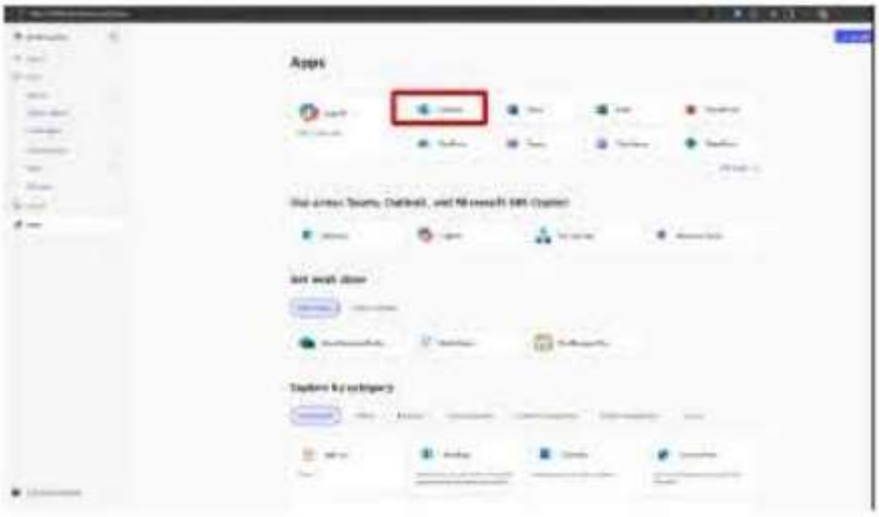
Login to office.com



Accessing Office Tools Online

1. **Microsoft Word Online**

- Click the Word icon from the dashboard.



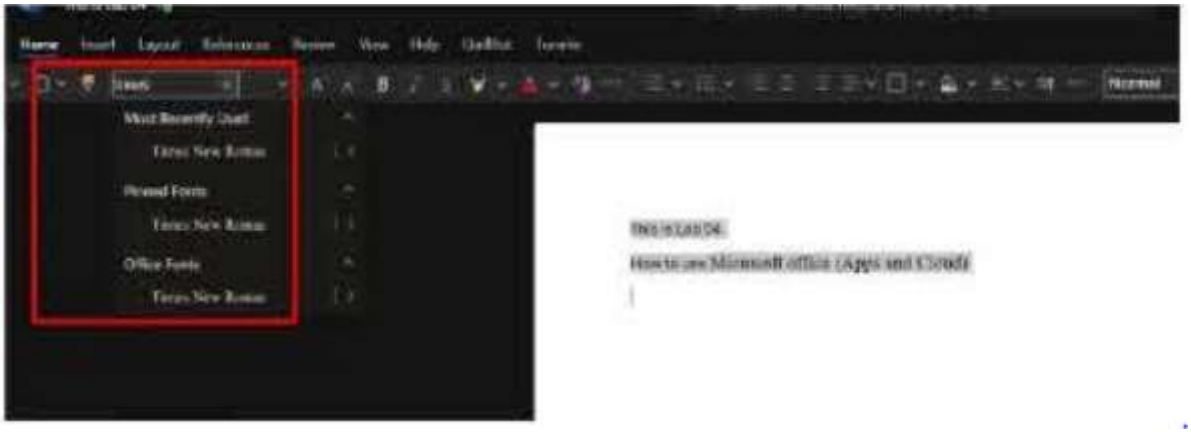
- Select New Blank Document or choose a template.



- Type, format, and edit text similarly to the desktop version.
- Documents are automatically saved in OneDrive.



- Change font style and size from the upper ribbon options.

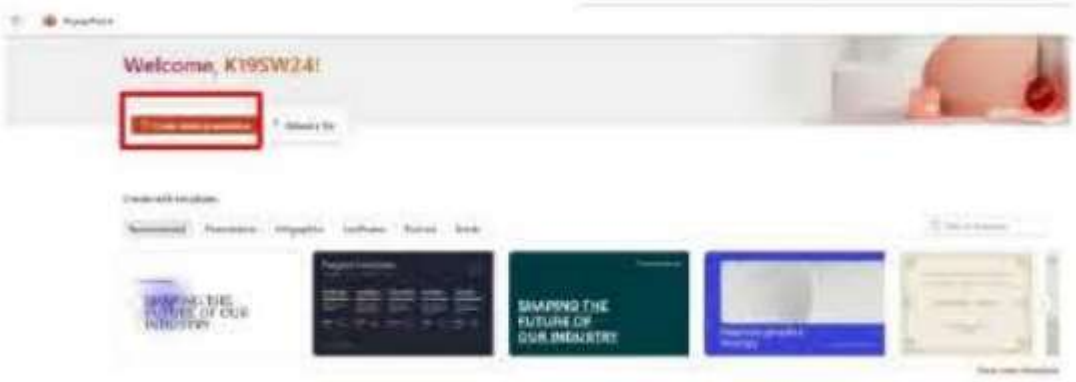


- You can choose ribbon style to ordinary



2. Microsoft PowerPoint Online

- Click the PowerPoint icon from the dashboard.
- Select New Blank Presentation or choose a design template.

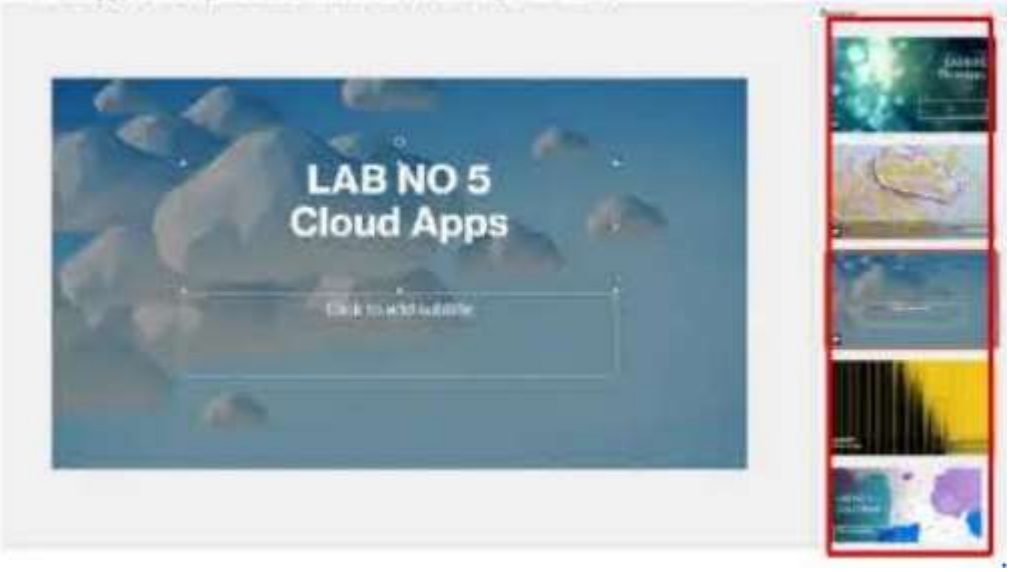


- Create slides with text, images, and animations.

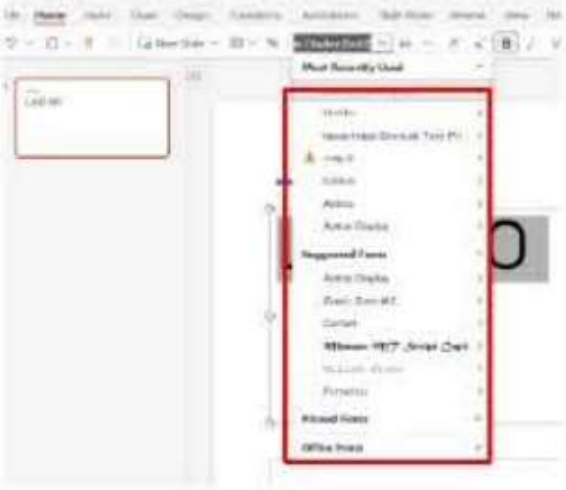


- Use the AI suggested Templates from the Designer option at the right side.

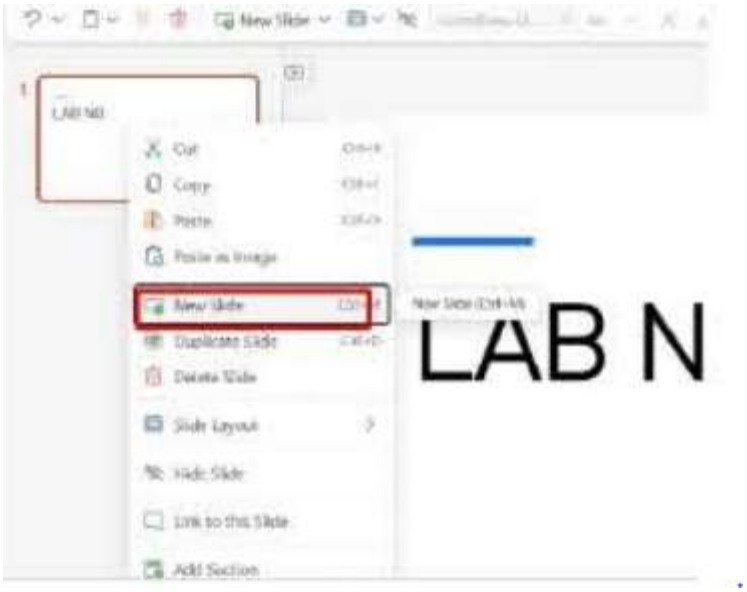
(AI suggest templated based on the text of the slide)



- Choose multiple options from the Ribbon and Header menu. Like font style, font size and other options from the menu.



- Create multiple slides with the menu option or follow the shortcut (Ctrl+M).



3. Microsoft Excel Online

- Click the Excel icon from the dashboard.
- Create and edit workbooks in the browser.
- Use tables, formulas, charts, and formatting tools.
- Collaborate with classmates by sharing the workbook link



Benefits of Office Online

- Accessibility: Work from anywhere, on any device.
- Collaboration: Share documents and edit simultaneously.
- Cloud storage: Files are saved automatically in OneDrive.
- Free to use: With a Microsoft account, students can use core apps at no cost.

Lab Tasks 05

Q1: Login to Office Online

- Open www.office.com and log in with your Microsoft account.
- Take a screenshot of your dashboard with your individual student name visible.

Q2: Create a Document in Word Online

- Open Word Online and create a document titled “My First Cloud Document”.
- Type a short paragraph about the advantages of cloud-based applications.

Q3: Design a Presentation in PowerPoint Online

- Create a 3-slide presentation about “The Benefits of Microsoft Office Online”.
- Use at least one image and a design theme.

Q4: Use Excel Online

- Open Excel Online and prepare a simple table showing your weekly study schedule.
- Apply bold headings and insert a chart for visualization.

Q5: Send an Email via Outlook Online

- Compose and send an email to your lab instructor with the subject “Lab 05 Completed”.
- Attach the Word and PowerPoint files you created.

Note: Upload screenshots or share links for each completed task as instructed.

Lab 06: Python Programming Fundamentals

Objective

The objective of this lab is to introduce students to the fundamental concepts of programming in Python. By practicing these concepts, students will learn how to declare variables, understand different data types, interact with users through input and output, use conditional statements for decision-making, and apply loops for repetition. This forms the foundation of programming that will help in solving larger problems.

Software Used

- Hardware: Desktop PC
- Software Tool: Online Compiler or Visual Studio Code

Introduction

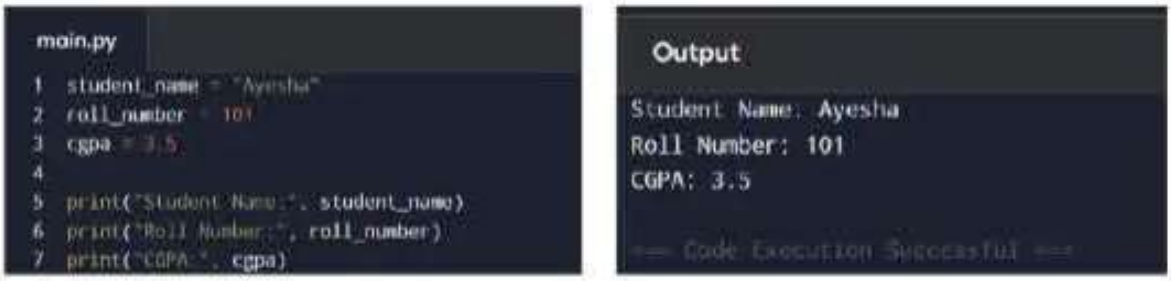
Programming is the art of giving instructions to a computer to perform specific tasks. Python is widely considered one of the best languages for beginners due to its simple syntax and powerful features. Before building advanced programs, we need to understand the basic building blocks of programming: variables, data types, input and output, conditionals, and loops. This lab will guide students through each of these topics with examples and exercises.

1. Variables

A variable is like a container that holds data. When we want to use a value multiple times in a program, we store it in a variable with a meaningful name. In Python, variables are created as soon as we assign a value to them. Unlike some languages, we do not need to declare the type first.
Variable names must follow these rules:

- Start with a letter or underscore.
- Cannot start with a number.
- Should not contain spaces or special characters except underscore.
- Should be descriptive.

Example Code



2. Data Types

Every value in Python has a specific type that defines what operations can be performed on it. Data types help us differentiate between numbers, text, and logical values. Python provides several built-in data types that we commonly use in programming.

Main Data Types in Python:

- **int**: Whole numbers such as 10 and -5.
- **float**: Decimal numbers such as 3.14 and -2.5.
- **str**: Text or characters such as “Hello”.
- **bool**: Logical values True or False.

Example Code



3. Input and Output

For any program to be useful, it should interact with the user. Python allows input from users using the input () function and displays output using the print () function. The input () function always returns data as a string, so when numbers are required, convert them using int () or float ().



Example Code

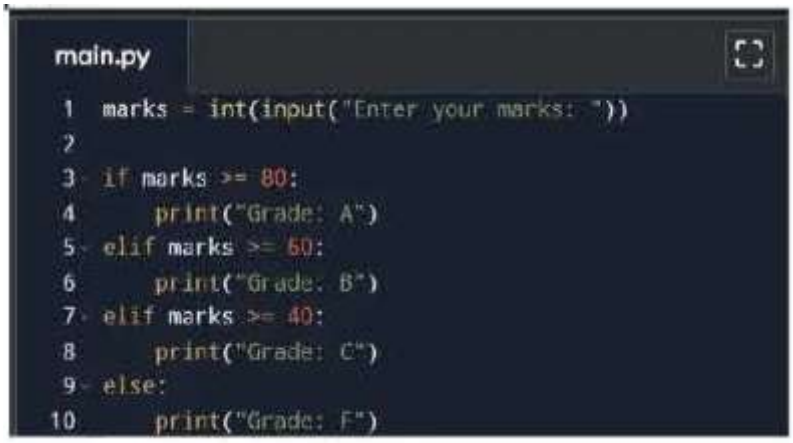


4. Conditional Statements

A conditional statement checks whether a certain condition is true or false. If the condition is true, the program executes one block of code. If condition is false, the program can either do nothing or execute a different block of code. This allow a program to behave differently in different situations instead of always following the same instructions.

- if → Executes code when condition is true.
- if-else → chooses between two possibilities.
- If-elif-else → Handles multiple conditions.

Example Code





5. Loops

A **for loop** is useful when the number of repetitions is fixed. For example, if we want to print numbers from 1 to 10, for loop is ideal because the number of repetitions are fixed.

A **while loop** is used when the number of repetitions is not fixed and depends on a condition. The loop continues as long as the condition is true. For example, if we want a program to keep asking the user for input until they type 'exit', we use a while loop because we do not know beforehand how many times the user will give input.

Note: Indentation (the space at the beginning of a line) in Python is essential. Code inside a loop must be indented to show it belongs to the loop. Also, with while loops, ensure the condition eventually becomes false to avoid an infinite loop.

For Loop Example

```
main.py
1 for i in range(1, 6):
2     print("Number:", i)
3
```

```
Number: 1
Number: 2
Number: 3
Number: 4
Number: 5
```

While Loop Example

```
main.py
1 count = 1
2 while count <= 5:
3     print("Number:", count)
4     count += 1
```

```
Number: 1
Number: 2
Number: 3
Number: 4
Number: 5
```

Lab Task 06

Q1: Write a program to store and print your name, roll number, and favorite subject.

Q2: Write a program to take two numbers and display their sum, difference, product, and division.

Q3: Write a program to check whether a number is even or odd.

Q4: Write a program to print the multiplication table of a number entered by the user.

Q5: Write a program that keeps asking for input until the user types "exit".

Q6: Write a program that prints only even numbers from 1 to 100.